Artificial Intelligence answer key (version 1.0.0)

Private grading material. Each entry: question ID, its page in the question collection, the correct choice, why it is right, why each other choice is wrong, whether arithmetic and code examples were checked by code, and the sources for the set.

S1-M-Q1 · question page 2

AI.BASICS.01 · Describe AI as a broad term for systems that do thinking-like tasks

Which description fits this course's meaning of AI?

Answer: D. A broad name for computer systems that do tasks linked with human thinking

AI is an umbrella term for many kinds of systems that do thinking-like tasks.

- ✗ A. Any computer program at all, including a basic pocket calculator: AI refers to tasks like recognizing, predicting, recommending and generating
- ✗ B. A single website that answers questions in a chat window: chatbots are only one kind of AI
- ✗ C. One specific robot that thinks in exactly the same way as a person: AI is a broad group, not one robot

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-M-Q2 · question page 3

AI.BASICS.01 · Describe AI as a broad term for systems that do thinking-like tasks

A phone groups photos by the faces in them, a music app suggests songs, and a chatbot writes a poem. Which statement is accurate?

Answer: C. All three can be called AI, but they do different jobs

Recognizing faces, recommending songs and writing text are all AI tasks, done by different systems.

- ✗ A. All three must run on the very same AI system inside: different tools work differently

✗ B. Only the chatbot counts as AI, because it writes words: recognizing and recommending are AI tasks too

✗ D. None of them counts as AI, since each does only one task: doing one task does not stop a system being AI

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-M-Q3 · question page 4

AI.BASICS.01 · Describe AI as a broad term for systems that do thinking-like tasks

A product box says "Now with AI!" What does that label tell you by itself?

Answer: B. Only that some feature uses AI, not how well it works

A label names a category; it does not describe quality or abilities.

✗ A. That the product now thinks the way a person thinks: the label says nothing about human-like thinking

✗ C. That it is more accurate than every older version: a label is not evidence of accuracy

✗ D. That every feature in the product learned from data: the label may describe just one feature

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-M-Q4 · question page 5

AI.BASICS.01 · Describe AI as a broad term for systems that do thinking-like tasks

Which three questions does the passage suggest asking about any AI tool?

Answer: A. What task it does, what it takes in, and what it gives back

Task, input and output describe what a tool actually does.

✗ B. How popular it is, how new it is, and who recommends it: popularity does not explain the task

✗ C. Who invented it, what year it came out, and what it costs: these do not

explain what the tool does

✗ D. Whether it has a robot body, a voice and a face: most AI has none of these

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-M-Q5 · question page 6

AI.BASICS.01 · Describe AI as a broad term for systems that do thinking-like tasks

A spam filter moves suspicious emails into a junk folder. Which AI task is it doing?

Answer: D. Sorting each email into a category

Deciding "junk" or "not junk" is sorting into categories, a recognition task.

✗ A. Recommending new contacts to add: it does not suggest contacts

✗ B. Writing new emails for the user: it does not write anything

✗ C. Translating emails into other languages: it does not translate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-B1-Q1 · question page 7

AI.BASICS.01 · See "AI" as a broad label, like the word "vehicle"

Why is "vehicle" a good comparison for "AI" in the passage?

Answer: C. Both words name a broad group of different things

Each word covers many different things that work differently.

✗ A. Both words describe ways of moving people around: AI does not move people

✗ B. Both words describe one machine with many parts: both name groups, not one machine

✗ D. Both words describe machines that have engines: many AI systems are not machines with engines

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S1-B1-Q2 · question page 8

AI.BASICS.01 · See "AI" as a broad label, like the word "vehicle"

Knowing that a tool is "AI" is like knowing that something is "a vehicle." What does that alone tell you?

Answer: A. Very little about what it can actually do

A broad label leaves out the details that matter.

- ✗ B. That it can do any job that a person can: the label promises no such thing
- ✗ C. Exactly how accurate and how fast it is: the label gives no measurements
- ✗ D. That it is safe to use for any purpose: the label says nothing about safety

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-B1-Q3 · question page 9

AI.BASICS.01 · See "AI" as a broad label, like the word "vehicle"

Which question tells you more about an AI tool?

Answer: B. What job does it do?

Asking about its job tells you what it is for.

- ✗ A. Is it labeled "AI"?: the label is broad
- ✗ C. Is it a new release?: newness is not ability
- ✗ D. Is its logo modern?: appearance is not ability

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-B1-Q4 · question page 10

AI.BASICS.01 · See "AI" as a broad label, like the word "vehicle"

A bus and a bicycle are both vehicles. A chatbot and a photo-tagging tool are both what?

Answer: A. AI systems that do different jobs

Like vehicles, AI systems share a label but differ.

- ✗ B. chatbots that only write text: a photo tagger is not a chatbot
- ✗ C. tools that cannot be called AI: both can be called AI
- ✗ D. the same AI system with two names: they are different systems

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-B1-Q5 · question page 11

AI.BASICS.01 · See "AI" as a broad label, like the word "vehicle"

Which task fits the passage's list of AI tasks (recognizing, predicting, recommending, generating)?

Answer: C. Suggesting a movie you might enjoy

Suggesting a movie is recommending.

- ✗ A. Printing the page you selected: an ordinary command, not one of the listed tasks
- ✗ B. Turning the screen brightness up: an ordinary command, not one of the listed tasks
- ✗ D. Saving a file when you click Save: an ordinary command, not one of the listed tasks

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F1-Q1 · question page 12

AI.BASICS.01 · Apply the broad meaning of AI to real claims

Lena's school announces an "AI tutor." What is the most useful first question to ask?

Answer: B. What exactly will it help students do?

Start with the task: what it does for students.

- ✗ A. Is it the same AI as every chatbot?: AI tools differ
- ✗ C. Which company made its logo and colors?: branding does not describe the task
- ✗ D. Is it smarter than all the teachers?: a vague comparison, not a description of the task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F1-Q2 · question page 13

AI.BASICS.01 · Apply the broad meaning of AI to real claims

A store's website says its prices are "set by AI." Which statement is accurate?

Answer: D. Some system helps set prices; the label alone does not say how

The label tells you a system is involved, not how it works or how well.

- ✗ A. A robot employee chooses each price by hand: AI here means software, not a robot worker
- ✗ B. A chatbot writes out every price for the store: pricing AI is usually not a chatbot
- ✗ C. The prices must be the lowest prices available anywhere: the label says nothing about low prices

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F1-Q3 · question page 14

AI.BASICS.01 · Apply the broad meaning of AI to real claims

Marcus says, "My map app isn't AI because it doesn't talk to me." What is wrong with his reasoning?

Answer: A. AI also includes systems that predict, like travel times

Predicting travel time is an AI task even without conversation.

✗ B. Map apps are never allowed to use any kind of AI: many map apps do use predictions

✗ C. AI means a robot, so an app can never qualify: AI usually means software

✗ D. Nothing; only talking systems can be called AI at all: AI is broader than talking systems

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F1-Q4 · question page 15

AI.BASICS.01 · Apply the broad meaning of AI to real claims

A job ad asks for "experience with AI." What should an applicant clarify?

Answer: D. Which AI tools or tasks the job actually involves

Because "AI" is broad, the specific tools and tasks matter.

✗ A. Whether the job needs building a human mind: unrealistic and not what such ads mean

✗ B. Whether the employer owns a humanlike robot: AI work rarely involves robots

✗ C. Whether AI is one program that everyone uses: AI is not one program

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F1-Q5 · question page 16

AI.BASICS.01 · Apply the broad meaning of AI to real claims

Two friends argue about whether a smart thermostat that learns your schedule is AI. Which response fits the passage best?

Answer: C. It can be called AI; what matters is the task it performs

Learning a schedule to predict settings is an AI task; the useful question is what

it does.

- ✗ A. It counts as AI only if it is as smart as a person: AI systems are usually narrow
- ✗ B. It cannot be AI because it has no screen: a screen is not required
- ✗ D. It counts as AI only if it can hold a conversation: conversation is not required

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F2-Q1 · question page 17

AI.BASICS.01 · Look past an "AI" label when a decision matters

A budgeting app calls itself "AI-powered," but it only applies one fixed rule: save 10 percent of each deposit. What is the best conclusion?

Answer: B. The label may be marketing; a fixed rule is doing the work

Labels are used loosely; look at what the system actually does.

- ✗ A. It is more reliable because it says "AI-powered": a label is not evidence
- ✗ C. It must be learning from millions of other users' data: nothing suggests learning
- ✗ D. It cannot work, since it is not "real" AI: a fixed rule can work fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F2-Q2 · question page 18

AI.BASICS.01 · Look past an "AI" label when a decision matters

Two résumé-screening tools are both described as "AI-powered." What should a hiring manager compare?

Answer: A. What each checks, how it was tested, and who reviews results

Quality and risk come from what the tool does and how it was checked.

- ✗ B. Nothing, since the same label means the same quality: same label,

possibly very different quality

✗ C. Which one has the newer logo and website: design says nothing about quality

✗ D. Which one uses the word "AI" more often: word count says nothing about quality

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F2-Q3 · question page 19

AI.BASICS.01 · Look past an "AI" label when a decision matters

A bank says AI helps decide loan applications. Which question looks past the label?

Answer: C. What information is used, and can a person review it?

Ask about inputs, testing and human review.

✗ A. Does the AI have a friendly name and avatar?: irrelevant to how decisions are made

✗ B. Is this year's AI newer than last year's AI?: newness is not a check

✗ D. Is it the same AI that powers popular chatbots?: does not reveal how decisions are made

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F2-Q4 · question page 20

AI.BASICS.01 · Look past an "AI" label when a decision matters

A photo app's background-blur feature never mentions AI, but it learned from many example photos. Which statement is accurate?

Answer: D. It may use AI even though it does not say so

Some tools use learned patterns without the label.

✗ A. It cannot use AI, because AI is never mentioned: labels are used loosely in both directions

✗ B. It must secretly be a chatbot underneath: learning from photos does not make it a chatbot

✗ C. Only features that are advertised can learn from data: unadvertised features can learn too

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S1-F2-Q5 · question page 21

AI.BASICS.01 · Look past an "AI" label when a decision matters

A news story says a city "used AI" to plan bus routes. What would you need to know before judging the plan?

Answer: B. What the system did, what data it used, and how it was checked

A label starts the questions; the task, data and checks answer them.

✗ A. Whether the city also uses AI for its parking meters: unrelated systems

✗ C. Whether the headline used the word "AI" in large type: headline wording is not evidence

✗ D. Whether AI is a popular topic in the news this year: popularity is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-M-Q1 · question page 23

AI.BASICS.02 · Tell written rules from learned patterns

A checkout page gives free shipping whenever an order is over 50 dollars. How was this behavior most likely produced?

Answer: C. A person wrote a rule for it

A fixed "if over 50, then free" step is a written rule.

✗ A. It learned from many past orders: a fixed threshold is a written rule

✗ B. It copied what a chatbot suggested: nothing suggests a chatbot is involved

✗ D. It guessed the amount at random: the result is consistent, not random

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-M-Q2 · question page 24

AI.BASICS.02 · Tell written rules from learned patterns

A spam filter was shown thousands of emails labeled "spam" or "not spam" and now sorts new emails. Where does its behavior mainly come from?

Answer: D. Patterns learned from the labeled examples

Its behavior was learned from labeled examples.

✗ A. A list of every spam email ever sent: it generalizes from examples instead

✗ B. The user approving each email by hand first: the filter sorts automatically

✗ C. A rule written for every possible email: no one could write a rule for every email

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-M-Q3 · question page 25

AI.BASICS.02 · Tell written rules from learned patterns

Which is an advantage of rule-based software?

Answer: A. You can read the rule to see why it acted

Written rules are predictable and readable.

✗ B. It never needs anyone to update it over time: rules may need updating when needs change

✗ C. It handles any situation, even ones nobody planned: rules miss unplanned cases

✗ D. It improves on its own as it sees more data: rules change only when

someone edits them

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-M-Q4 · question page 26

AI.BASICS.02 · Tell written rules from learned patterns

Which is a common weakness of a system that learned from examples?

Answer: B. It can make surprising mistakes on unusual inputs

Learned behavior can fail in unexpected ways.

✗ A. It can never handle messy or varied input: learned systems often handle variety well

✗ C. It must be rewritten by hand for each new case: it generalizes from examples instead

✗ D. It can follow only one fixed rule at a time: that describes simple rule software

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-M-Q5 · question page 27

AI.BASICS.02 · Tell written rules from learned patterns

A thermostat turns the heat on at exactly 6:00 a.m. every weekday. Which describes it?

Answer: C. A written rule that runs on a schedule

A fixed schedule is a written rule.

✗ A. A chatbot predicting the weather each day: nothing here is a chatbot or a forecast

✗ B. A model trained on thousands of houses: no training is needed for a fixed time

✗ D. A learned pattern from the family's habits: a fixed time set in advance is a rule

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-B1-Q1 · question page 28

AI.BASICS.02 · Compare a recipe with learning by tasting

Which cook is like rule-based software?

Answer: A. The one who follows the recipe exactly

A recipe is a set of written rules.

- ✗ B. The one who learned by tasting many soups: that cook learned patterns
- ✗ C. The one who refuses to cook new dishes: not described in the passage
- ✗ D. The one who changes the recipe at random: neither cook works at random

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-B1-Q2 · question page 29

AI.BASICS.02 · Compare a recipe with learning by tasting

Why can the recipe cook explain her soup easily?

Answer: B. She can point to the written steps

Written rules can be read back as the explanation.

- ✗ A. She tasted hundreds of soups first: that is the other cook
- ✗ C. She guesses differently each time: she follows fixed steps
- ✗ D. She asks the diners to decide: not in the passage

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-B1-Q3 · question page 30

AI.BASICS.02 · Compare a recipe with learning by tasting

What can the tasting cook do that the recipe cook cannot?

Answer: D. Handle ingredients the recipe never mentions

Learning from many examples helps with new situations.

- ✗ A. Follow a written recipe exactly, step by step: that is the recipe cook's method
- ✗ B. Get every new dish right every time: she sometimes gets new dishes wrong
- ✗ C. Always explain each choice exactly: she often cannot explain exactly

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-B1-Q4 · question page 31

AI.BASICS.02 · Compare a recipe with learning by tasting

What is a weakness of the tasting cook?

Answer: C. She sometimes gets a new dish wrong

Learned skill can fail in surprising ways.

- ✗ A. She cannot handle any new ingredient: she handles new ingredients well
- ✗ B. She needs the recipe for every step: she does not use a recipe
- ✗ D. She never adjusts what she does: she adjusts constantly

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-B1-Q5 · question page 32

AI.BASICS.02 · Compare a recipe with learning by tasting

A system learned to recognize birds from thousands of labeled photos. Which cook is it like?

Answer: D. The tasting cook

Learning from examples matches the tasting cook.

- ✗ A. Neither cook: it matches the tasting cook

✗ B. Both cooks equally: it learned rather than followed rules

✗ C. The recipe cook: no one wrote bird rules

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F1-Q1 · question page 33

AI.BASICS.02 · Decide whether a task suits rules or learned patterns

A library wants to charge 25 cents per day for each overdue book. Which approach fits best?

Answer: A. A written rule, because the policy is exact and simple

An exact, simple policy is best written as a rule.

✗ B. A chatbot that guesses a fair fee for each reader: fees should be exact, not guessed

✗ C. A learned model trained on years of borrowing data: unnecessary for an exact policy

✗ D. A model that learns each librarian's fee habits: the policy is already fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F1-Q2 · question page 34

AI.BASICS.02 · Decide whether a task suits rules or learned patterns

An app must recognize handwritten numbers written by thousands of different people. Which approach fits best?

Answer: B. Learning patterns from many labeled examples

Messy, varied input is where learned patterns help.

✗ A. A fixed rule checking that every 7 has a line: handwriting varies too much

✗ C. Asking users to type each number instead: avoids the task rather than doing it

✗ D. One written rule for each person's handwriting: impossible to list every

style

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F1-Q3 · question page 35

AI.BASICS.02 · Decide whether a task suits rules or learned patterns

A school rule says any student absent 3 days in a row gets a call home. Which approach should run it?

Answer: C. A written rule that checks for 3 days in a row

An exact policy should run as a readable rule.

- ✗ A. A model that learns from past principals' choices: the school wants its exact rule followed
- ✗ B. A chatbot that decides case by case each morning: case-by-case guessing breaks the rule
- ✗ D. A model trained to guess which families want calls: the policy is already exact

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F1-Q4 · question page 36

AI.BASICS.02 · Decide whether a task suits rules or learned patterns

A photo service wants to suggest "beach" or "forest" tags for vacation pictures. Which approach fits best?

Answer: A. A model that learned from many labeled photos

Recognizing scenes from pixels needs learned patterns.

- ✗ B. A rule that marks every photo taken in July "beach": dates do not show scenes
- ✗ C. A rule that checks whether a file name says "beach": file names are often unrelated
- ✗ D. Asking a person to tag every photo by hand forever: possible but not the

AI approach asked about

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F1-Q5 · question page 37

AI.BASICS.02 · Decide whether a task suits rules or learned patterns

Which situation most needs the predictability of written rules?

Answer: D. Calculating sales tax that the law sets as an exact rate

An exact legal rate should be computed by a rule.

- ✗ A. Sorting thousands of customer comments by topic: varied text suits learned patterns
- ✗ B. Recognizing which animal appears in a photo: varied images suit learned patterns
- ✗ C. Suggesting songs a listener might enjoy next: preferences suit learned patterns

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F2-Q1 · question page 38

AI.BASICS.02 · Find which part failed in a system that mixes rules and learning

A bank's model flags a normal grocery purchase as unusual, and a written rule then holds the payment. Which part made the first mistake?

Answer: B. The learned model that flagged the purchase

The error began with the learned flag; the rule only followed it.

- ✗ A. The grocery store's card reader: nothing suggests the reader failed
- ✗ C. The written rule that held the payment: it did what it was written to do
- ✗ D. The customer who bought the groceries: the purchase was normal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F2-Q2 · question page 39

AI.BASICS.02 · Find which part failed in a system that mixes rules and learning

A written rule says "hold flagged payments over 100 dollars," but the bank meant 1,000. How can this be fixed?

Answer: D. Rewrite the rule with the correct amount

A wrong written rule is found and rewritten.

- ✗ A. Retrain the model on more transactions: the model is not the problem
- ✗ B. Wait for the rule to learn the right amount: written rules do not learn
- ✗ C. Remove the learned model from the system: the model was not at fault

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F2-Q3 · question page 40

AI.BASICS.02 · Find which part failed in a system that mixes rules and learning

A learned model keeps misreading a new style of receipt. What is the usual way to improve it?

Answer: C. Retrain it with examples of the new receipts

Learned patterns usually change through retraining with better examples.

- ✗ A. Make the receipts look like old ones: unrealistic and avoids the problem
- ✗ B. Tell it the correct answer once in a chat: a chat does not retrain the model
- ✗ D. Edit one line of its rules by hand: learned patterns are not written lines

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F2-Q4 · question page 41

AI.BASICS.02 · Find which part failed in a system that mixes rules and learning

Why might a designer use a written rule for the final payment decision even when a model flags cases?

Answer: B. Rules give predictable results where it matters most

Combining them uses learning for variety and rules for predictability.

- ✗ A. Rules handle every possible case automatically: rules only cover listed cases
- ✗ C. Rules never need to be checked or updated: rules still need review
- ✗ D. Rules can recognize unusual payments by themselves: recognition is the model's job

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S2-F2-Q5 · question page 42

AI.BASICS.02 · Find which part failed in a system that mixes rules and learning

A customer asks why their payment was held. Which part is easier to explain?

Answer: A. The written rule about which flagged payments are held

The rule can be read; the learned flag may be harder to explain.

- ✗ B. Both parts are always equally easy to explain: learned patterns are usually harder
- ✗ C. The exact reason the model flagged the payment: learned patterns can be hard to explain
- ✗ D. Neither part can ever be explained at all: the rule can be read

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-M-Q1 · question page 44

AI.BASICS.03 · Explain how AI, machine learning and deep learning nest

How are AI, machine learning and deep learning related?

Answer: C. Deep learning is part of machine learning, which is part of AI

The groups nest: deep learning inside machine learning inside AI.

✗ A. They are three different words for exactly the same thing: each is a narrower group

✗ B. AI is part of deep learning, which is part of machine learning: the nesting is reversed

✗ D. Machine learning and deep learning are unrelated to AI: both are kinds of AI

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-M-Q2 · question page 45

AI.BASICS.03 · Explain how AI, machine learning and deep learning nest

What does "deep" mean in "deep learning"?

Answer: B. The network has many layers

"Deep" counts layers in the network.

✗ A. The training data is kept hidden: not what "deep" refers to

✗ C. The system understands ideas deeply: "deep" is about layers, not understanding

✗ D. The model is very old and stable: not what "deep" refers to

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-M-Q3 · question page 46

AI.BASICS.03 · Explain how AI, machine learning and deep learning nest

What is the key idea of machine learning?

Answer: D. A system improves at a task by learning patterns from data

Machine learning learns from data instead of only following written rules.

✗ A. A person writes a separate rule for every possible case: that describes rule-based software

✗ B. A program stores every answer and repeats it: learning patterns is not

memorizing every answer

✗ C. A computer copies human brain cells into a chip: not what machine learning means

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-M-Q4 · question page 47

AI.BASICS.03 · Explain how AI, machine learning and deep learning nest

In a neural network, what do the layers do?

Answer: A. Pass numbers along, turning simple features into complex ones

Layers transform numbers step by step.

✗ B. Connect the computer to the internet to look up answers: layers are math steps, not connections

✗ C. Store a separate copy of every training example: layers transform numbers, not stored copies

✗ D. Hold the written rules that a programmer typed: learned weights, not typed rules

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-M-Q5 · question page 48

AI.BASICS.03 · Explain how AI, machine learning and deep learning nest

Which statement is accurate?

Answer: C. Some AI uses no machine learning at all

Rule-based systems can count as AI without machine learning.

✗ A. All AI is deep learning by definition: deep learning is only part of AI

✗ B. Deep learning is a separate field from AI: it is inside AI

✗ D. All machine learning uses deep neural networks: many methods are not deep learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-B1-Q1 · question page 49

AI.BASICS.03 · Picture nested boxes

Which is the biggest box?

Answer: A. Artificial intelligence

AI is the outer box.

- ✗ B. Machine learning box: the middle box
- ✗ C. Deep learning: the smallest box
- ✗ D. Neural network: a tool used in deep learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-B1-Q2 · question page 50

AI.BASICS.03 · Picture nested boxes

Where does deep learning sit?

Answer: D. Inside machine learning

Deep learning is the smallest, innermost box.

- ✗ A. Beside AI, not in it: it is inside AI
- ✗ B. Around machine learning: it is inside, not around
- ✗ C. Outside the AI box: it is inside AI

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-B1-Q3 · question page 51

AI.BASICS.03 · Picture nested boxes

A deep learning system is in which boxes?

Answer: B. All three boxes

Something in the smallest box is in every box around it.

- ✗ A. None of the boxes: it is in all of them
- ✗ C. Only the AI box: it is also in the inner boxes
- ✗ D. Only the deep learning box: inner boxes are also inside the outer ones

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-B1-Q4 · question page 52

AI.BASICS.03 · Picture nested boxes

A chess program built only from hand-written rules is in which box?

Answer: D. The AI box only

Rules alone can be AI without machine learning.

- ✗ A. The machine learning box: it does not learn
- ✗ B. The deep learning box: it does not learn
- ✗ C. No box at all: it does a thinking-like task, so it is AI

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-B1-Q5 · question page 53

AI.BASICS.03 · Picture nested boxes

What puts something in the machine learning box?

Answer: A. It learns patterns from data

Learning from data defines machine learning.

- ✗ B. It has many hand-written rules: that does not involve learning
- ✗ C. It runs on a fast computer: speed is not learning
- ✗ D. It has a robot body: a body is not required

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F1-Q1 · question page 54

AI.BASICS.03 · Classify systems as AI, machine learning or deep learning

A tool sorts emails using only rules a person wrote, such as "if the subject contains FREE, mark as junk." Which label fits best?

Answer: C. AI built from rules, not machine learning

Hand-written rules can be AI without machine learning.

- ✗ A. Deep learning with many network layers: no learning is involved
- ✗ B. Not AI, because it cannot write emails: writing is not required to be AI
- ✗ D. Machine learning trained on labeled emails: it only follows written rules

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F1-Q2 · question page 55

AI.BASICS.03 · Classify systems as AI, machine learning or deep learning

An app recognizes plant species from photos using a neural network with dozens of layers. Which label is most specific?

Answer: B. Deep learning

A many-layered neural network is deep learning, the most specific label.

- ✗ A. Not AI, since it only looks at pictures: recognition is an AI task
- ✗ C. Machine learning only, not deep learning: dozens of layers make it deep learning
- ✗ D. Rule-based AI without learning: it learned from data

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F1-Q3 · question page 56

AI.BASICS.03 · Classify systems as AI, machine learning or deep learning

A shop predicts next week's sales from past weekly sales using a simple formula fitted to the data, with no neural network. Which label fits?

Answer: A. Machine learning that is not deep learning

Fitting a simple model to data is machine learning without deep learning.

- ✗ B. Deep learning, since it learns from data: no many-layered network is used
- ✗ C. Rule-based AI written by hand: the formula was fitted to data
- ✗ D. Not AI, since it predicts numbers: prediction can be machine learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F1-Q4 · question page 57

AI.BASICS.03 · Classify systems as AI, machine learning or deep learning

Someone says, "It uses deep learning, so it truly understands." What is the best response?

Answer: D. "Deep" means many layers, not understanding

The word "deep" describes structure, not understanding.

- ✗ A. That is right; deep systems understand meaning: "deep" refers to layers
- ✗ B. Deep learning cannot be used for language tasks: it often is
- ✗ C. That is right; neural networks are real brains: they are math, not brains

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F1-Q5 · question page 58

AI.BASICS.03 · Classify systems as AI, machine learning or deep learning

Which example sits in the machine learning box but not the deep learning box?

Answer: C. A model fitted to past data with no neural network

Machine learning includes methods without deep neural networks.

✗ A. A list of rules typed in by a programmer by hand: that is not machine learning

✗ B. A calculator adding two numbers: that is not machine learning

✗ D. A neural network with forty layers: that is deep learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F2-Q1 · question page 59

AI.BASICS.03 · Read claims about neural networks and deep learning carefully

An ad says its "neural network works just like a human brain." What is the problem with this claim?

Answer: B. Neural networks are math inspired by a biology word, not brains

The brain comparison is a loose analogy.

✗ A. Neural networks cannot be used in any products: they are used widely

✗ C. Neural networks are physically built from brain cells: they are software and math

✗ D. Nothing; that comparison is scientifically exact: the comparison is loose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F2-Q2 · question page 60

AI.BASICS.03 · Read claims about neural networks and deep learning carefully

A small business has 200 rows of tidy sales numbers and wants a simple forecast it can explain. What does the passage suggest?

Answer: C. A simpler method may work as well and be easier to explain

Choose the approach that fits the task, data and need to explain.

✗ A. Deep learning is always required for any forecast: not for small, tidy problems

✗ B. Forecasting is impossible without a neural network: simple methods can forecast

✗ D. More layers always produce more accurate forecasts: not on small, tidy data

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F2-Q3 · question page 61

AI.BASICS.03 · Read claims about neural networks and deep learning carefully

Which is a common cost of deep learning compared with simpler methods?

Answer: A. It often needs more data and computing power

Deep learning often needs more data and computing.

✗ B. It must be written by hand line by line: it is trained, not hand-written
✗ C. It can only handle a few numbers at a time: it handles large inputs
✗ D. It cannot learn from examples at all: it learns from examples

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F2-Q4 · question page 62

AI.BASICS.03 · Read claims about neural networks and deep learning carefully

A headline says, "Deep AI now thinks deeply about your questions." What should a careful reader conclude?

Answer: D. The headline mixes up "deep" (layers) with deep thought

The headline uses "deep" loosely.

✗ A. The system has been proven to reflect like a person: nothing proves that
✗ B. "Deep" means the system was trained for a long time: "deep" means many layers
✗ C. All AI is now deep learning, so it thinks: not all AI is deep learning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S3-F2-Q5 · question page 63

AI.BASICS.03 · Read claims about neural networks and deep learning carefully

A team must explain each decision to customers and has little data. Which consideration matters most when choosing an approach?

Answer: B. Whether the method can be explained with this data

Task, data and the need to explain should guide the choice.

- ✗ A. Whether the method is described as "deep": the label does not matter
- ✗ C. Whether competitors call their tools neural: competitors' labels are irrelevant
- ✗ D. Whether the method has the most layers possible: more layers can make explaining harder

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-M-Q1 · question page 65

AI.BASICS.04 · Classify an AI use as recommending, recognizing, forecasting or generating

A video site shows "Up next" suggestions based on what you watched. Which job is this?

Answer: D. Recommending

Suggesting items you might want is recommending.

- ✗ A. Recognizing: it is not putting input into categories
- ✗ B. Forecasting: it is not predicting a number
- ✗ C. Generating: it is not creating new content

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-M-Q2 · question page 66

AI.BASICS.04 · Classify an AI use as recommending, recognizing, forecasting or generating

A phone unlocks when it identifies its owner's face. Which job is this?

Answer: B. Recognizing

Identifying the face puts the input into a category.

- ✗ A. Recommending: it does not suggest anything
- ✗ C. Generating: it does not create content
- ✗ D. Forecasting: it does not predict a future value

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-M-Q3 · question page 67

AI.BASICS.04 · Classify an AI use as recommending, recognizing, forecasting or generating

A weather app predicts a 70 percent chance of rain tomorrow. Which job is this?

Answer: C. Forecasting

Predicting a future event is forecasting.

- ✗ A. Recommending: it suggests no items
- ✗ B. Recognizing: it is predicting, not sorting what is there
- ✗ D. Generating: a probability is a forecast, not new content

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-M-Q4 · question page 68

AI.BASICS.04 · Classify an AI use as recommending, recognizing, forecasting or generating

A tool writes a birthday poem from a short description. Which job is this?

Answer: A. Generating

Creating new text is generating.

- ✗ B. Recognizing: it creates rather than categorizes
- ✗ C. Recommending: it suggests no existing items
- ✗ D. Forecasting: it predicts no future value

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-M-Q5 · question page 69

AI.BASICS.04 · Classify an AI use as recommending, recognizing, forecasting or generating

A voice assistant hears "What's the weather?" and then writes a spoken reply. Which jobs does it do?

Answer: C. Recognizing speech, then generating a reply

One app can chain jobs: recognize, then generate.

- ✗ A. Only forecasting what the user will ask next: it recognizes the actual question
- ✗ B. Only recommending a weather app to install: it recognizes and replies
- ✗ D. Only generating; no recognition is needed: it must first recognize the spoken words

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-B1-Q1 · question page 70

AI.BASICS.04 · Match each job to a familiar helper

The cashier knows a scanned item is a banana. Which job is that?

Answer: A. Recognizing

Identifying an item is recognizing.

- ✗ B. Generating: that is the sign painter
- ✗ C. Recommending: that is the clerk
- ✗ D. Forecasting: that is the manager

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-B1-Q2 · question page 71

AI.BASICS.04 · Match each job to a familiar helper

The manager predicts Saturday bread sales. Which job is that?

Answer: B. Forecasting

Predicting future sales is forecasting.

- ✗ A. Generating: no new content is made
- ✗ C. Recommending: nothing is suggested to a buyer
- ✗ D. Recognizing: nothing is being identified

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-B1-Q3 · question page 72

AI.BASICS.04 · Match each job to a familiar helper

Which helper is generating?

Answer: D. The sign painter

Creating a new poster is generating.

- ✗ A. The manager: forecasts
- ✗ B. The cashier: recognizes
- ✗ C. The sales clerk: recommends

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-B1-Q4 · question page 73

AI.BASICS.04 · Match each job to a familiar helper

"People who bought this also liked that" matches which job?

Answer: C. Recommending

Suggesting items is recommending.

- ✗ A. Forecasting: no future number is predicted

✗ B. Generating: nothing new is created

✗ D. Recognizing: nothing is identified

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-B1-Q5 · question page 74

AI.BASICS.04 · Match each job to a familiar helper

An app predicts how many minutes your bus will take. Which shop helper does the same kind of job?

Answer: A. The manager

Predicting a number is forecasting, like the manager.

✗ B. The cashier: recognizes items

✗ C. The sales clerk: suggests products

✗ D. The sign painter: creates new posters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F1-Q1 · question page 75

AI.BASICS.04 · Name the AI job in real apps

A bank text says, "We blocked a payment that looked like fraud." Which job did the AI most likely do?

Answer: B. Recognizing the payment as likely fraud

Classifying a payment as fraud is recognizing.

✗ A. Forecasting next year's interest rates: unrelated to the block

✗ C. Recommending a product to the customer: nothing was suggested

✗ D. Generating a new payment for the customer: no new payment was made

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F1-Q2 · question page 76

AI.BASICS.04 · Name the AI job in real apps

A delivery app says your order will arrive in 25 minutes. Which job is this?

Answer: D. Forecasting

Predicting arrival time is forecasting.

- ✗ A. Recognizing: it predicts rather than identifies
- ✗ B. Generating: a time estimate is a forecast
- ✗ C. Recommending: it suggests no items

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F1-Q3 · question page 77

AI.BASICS.04 · Name the AI job in real apps

A design app makes three new logo drafts from a description. Which job is this?

Answer: C. Generating

Creating new images is generating.

- ✗ A. Recognizing: it creates new images
- ✗ B. Forecasting: it predicts no number
- ✗ D. Recommending: the logos are new, not existing items

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F1-Q4 · question page 78

AI.BASICS.04 · Name the AI job in real apps

A shopping site shows "Customers also bought" items. What could go wrong that fits this job?

Answer: A. It keeps suggesting more of the same, missing your real need

Recommenders can narrow suggestions to more of the same.

✗ B. It writes a false product description: a generating problem

✗ C. It mislabels your photo as a cat: a recognizing problem

✗ D. It miscounts the exact delivery time of your order: a forecasting problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F1-Q5 · question page 79

AI.BASICS.04 · Name the AI job in real apps

A camera app labels a photo of a husky as "wolf." Which job made the mistake?

Answer: B. Recognizing

Putting the photo in the wrong category is a recognition error.

✗ A. Recommending: it suggested nothing

✗ C. Forecasting: it predicted no future value

✗ D. Generating: it did not create content

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F2-Q1 · question page 80

AI.BASICS.04 · Use the job type to decide what to check

A voice app answers "Book a table for four" by booking a table for two. Speech was turned into text, then a booking was made. What should be checked first?

Answer: D. Whether the speech was recognized correctly

In a chain, an early recognition error spreads to later steps.

✗ A. Whether the booking page used AI at all: the question is which step failed

✗ B. Whether the restaurant forecast was wrong: no forecast was involved

✗ C. Whether the recommendations were too narrow: no recommendation was involved

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S4-F2-Q2 · question page 81

AI.BASICS.04 · Use the job type to decide what to check

A sales forecast built from three normal years badly misses a month when a new factory opened nearby. Why?

Answer: A. Forecasts can miss sudden changes not in past data

A change never seen in the data can surprise a forecast.

- ✗ B. Recommenders made the forecast too narrow: no recommender was involved
- ✗ C. Forecasts always ignore all past data: they rely on past data
- ✗ D. The forecast must have generated false text: forecasts predict numbers

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F2-Q3 · question page 82

AI.BASICS.04 · Use the job type to decide what to check

A tool generated a confident product description listing a feature the product does not have. What kind of failure is this?

Answer: B. A generator producing fluent but false content

Generators can write fluent content that is false.

- ✗ A. A forecaster missing a sudden change: nothing was forecast
- ✗ C. A recommender narrowing suggestions: nothing was recommended
- ✗ D. A recognizer putting the product in a category: nothing was categorized

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F2-Q4 · question page 83

AI.BASICS.04 · Use the job type to decide what to check

A music app plays only songs very similar to ones you already liked. What is the best fix to try?

Answer: D. Look for settings or searches that widen the suggestions

Recommenders can narrow choices; widen them on purpose.

- ✗ A. Retrain its speech recognition to understand your voice: the issue is recommendations
- ✗ B. Ask it to generate brand-new songs instead: avoids the recommendation problem
- ✗ C. Check its weather forecast against a website: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S4-F2-Q5 · question page 84

AI.BASICS.04 · Use the job type to decide what to check

A photo tool identifies plants well in a garden but poorly for rare desert plants. What does this suggest?

Answer: C. Recognizers struggle with unusual examples

Recognition errors increase on unusual input.

- ✗ A. Forecasts miss sudden changes in sales: nothing is forecast
- ✗ B. Generators invent fluent but false text: nothing is generated
- ✗ D. Recommenders repeat the same suggestions: nothing is recommended

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-M-Q1 · question page 86

AI.BASICS.05 · Decide whether a task needs prediction or generation

A school wants each incoming email marked "absence," "question" or "other." Which kind of AI use is this?

Answer: A. Predictive: it produces a label

Assigning a category is a predictive (classifying) use.

- ✗ B. Generative: it writes new emails: it labels emails; it does not write them
- ✗ C. Generative: it creates new categories each day: the categories are fixed labels
- ✗ D. Neither; labeling emails cannot use AI: labeling is a classic AI task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-M-Q2 · question page 87

AI.BASICS.05 · Decide whether a task needs prediction or generation

A student asks a tool for a short story about a lost robot. Which kind of use is this?

Answer: D. Generative: it produces new content

Writing a new story is generative.

- ✗ A. Predictive: it produces a label: a story is new content
- ✗ B. Neither; stories cannot be produced by AI: generating stories is a common use
- ✗ C. Predictive: it produces one number: a story is not a number

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-M-Q3 · question page 88

AI.BASICS.05 · Decide whether a task needs prediction or generation

Which output is typical of a predictive use?

Answer: C. "87 percent likely to be fraud"

A probability or label is a prediction.

- ✗ A. A paragraph of product description: generative
- ✗ B. A drawing of a mountain at sunset: generative
- ✗ D. A new poem about the ocean: generative

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-M-Q4 · question page 89

AI.BASICS.05 · Decide whether a task needs prediction or generation

What is a typical failure of generated content?

Answer: B. It looks polished but contains false or invented details

Generators can sound polished while being wrong.

- ✗ A. It returns a predicted number that is a bit too high: that is a predictive failure
- ✗ C. It refuses to produce any output at all: possible, but not the typical failure described
- ✗ D. It gives a single wrong yes or no label: that is a predictive failure

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-M-Q5 · question page 90

AI.BASICS.05 · Decide whether a task needs prediction or generation

A clinic wants to estimate how many patients will arrive tomorrow. Which kind of use fits?

Answer: D. Predictive: a number

An estimated count is a predictive output.

- ✗ A. Generative: a new image: they need a number
- ✗ B. Generative: a song about patients: they need a number
- ✗ C. Generative: a written story: they need a number

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-B1-Q1 · question page 91

AI.BASICS.05 · Tell a label or number from new content

Which output is a label?

Answer: B. "Approved"

"Approved" names a category.

- ✗ A. A photo of a sunset: new content
- ✗ C. A three-paragraph essay: new content
- ✗ D. A new song melody: new content

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-B1-Q2 · question page 92

AI.BASICS.05 · Tell a label or number from new content

Which output is a number?

Answer: A. "12 percent chance of rain"

A percentage measures a chance.

- ✗ B. "Spam": a label
- ✗ C. "A new drawing of clouds": new content
- ✗ D. "A poem about rain": new content

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-B1-Q3 · question page 93

AI.BASICS.05 · Tell a label or number from new content

Which output is new content?

Answer: C. A newly written joke

A newly written joke did not exist before.

- ✗ A. "Cat (label)": a label
- ✗ B. "Not spam": a label

✗ D. "About 42 minutes": a number

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-B1-Q4 · question page 94

AI.BASICS.05 · Tell a label or number from new content

"Spam (95 percent sure)" is which kind of output?

Answer: B. A label plus a number

It combines a category and a confidence number.

✗ A. An essay: it is a short label and score
✗ C. New content only, like an essay: nothing new was written
✗ D. A number only: it also has a label

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-B1-Q5 · question page 95

AI.BASICS.05 · Tell a label or number from new content

Which question checks a label?

Answer: C. "Which category is it?"

A label answers "which category?"

✗ A. "Does the song rhyme?": about content
✗ B. "How long is the story?": about content length
✗ D. "Is the drawing pretty?": about content

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F1-Q1 · question page 96

AI.BASICS.05 · Choose the right kind of AI use for a task

A store wants to know which product reviews are positive or negative. Which use fits best?

Answer: A. Predictive: label each review as positive or negative

Sorting reviews by sentiment is a labeling task.

- ✗ B. Generative: write a poem about the reviews: they need labels
- ✗ C. Generative: write new reviews for each product: they want to sort existing reviews
- ✗ D. Generative: create images of happy customers: unrelated to sorting reviews

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F1-Q2 · question page 97

AI.BASICS.05 · Choose the right kind of AI use for a task

A teacher wants five new practice questions about fractions. Which use fits best?

Answer: D. Generative: create new questions

Creating new questions is generative.

- ✗ A. Predictive: score one answer as right or wrong: not what was asked
- ✗ B. Predictive: label existing questions by topic: they need new questions
- ✗ C. Predictive: forecast next week's test scores: not what was asked

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F1-Q3 · question page 98

AI.BASICS.05 · Choose the right kind of AI use for a task

A tool generated five practice questions. What should the teacher check before using them?

Answer: C. That each question and answer is actually correct

Generated content can look polished but contain errors.

✗ A. That a label was predicted for each question: the risk is false content

✗ B. That the questions are as long as possible: length is not correctness

✗ D. Nothing, because generated text is checked already: generated content can be wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F1-Q4 · question page 99

AI.BASICS.05 · Choose the right kind of AI use for a task

A city wants an estimate of how many buses are needed at 8 a.m. Which use fits best?

Answer: B. Predictive: estimate a number

An estimated count is predictive.

✗ A. Generative: compose a bus jingle: they need a number

✗ C. Generative: draw a picture of a bus: they need a number

✗ D. Generative: write a story about commuters: they need a number

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F1-Q5 · question page 100

AI.BASICS.05 · Choose the right kind of AI use for a task

A volunteer group wants a friendly thank-you letter for donors. Which use fits best?

Answer: D. Generative: draft a new letter

Drafting a letter is generative; a person should still review it.

✗ A. Predictive: mark the letter "approved": nothing exists to approve yet

✗ B. Predictive: estimate next year's donations: they need a letter

✗ C. Predictive: label each donor by city: they need a letter

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F2-Q1 · question page 101

AI.BASICS.05 · Handle tasks that mix prediction and generation

A help desk tool labels a billing question as "shipping," then drafts a confident reply about delivery times. What went wrong?

Answer: A. The wrong label led to a confidently wrong draft

An early prediction error spread into the generated reply.

- ✗ B. The label was right, but the draft was too short: the label was wrong
- ✗ C. Generators can never write help desk replies: they can, with checking
- ✗ D. The draft was right; billing and shipping are the same: they are different topics

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F2-Q2 · question page 102

AI.BASICS.05 · Handle tasks that mix prediction and generation

A company needs a yes or no decision on each expense claim, with a score and a record it can audit. Which approach is easier to test?

Answer: B. A classifier that outputs a label and a score

A predictive output fits a yes or no decision with an audit trail.

- ✗ A. A generator that writes a paragraph about each claim: free text is harder to test
- ✗ C. A generator that writes a poem about expenses: unrelated
- ✗ D. A generator asked to "decide fairly" in prose: harder to test and audit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F2-Q3 · question page 103

AI.BASICS.05 · Handle tasks that mix prediction and generation

A generated travel guide says a museum is free on Mondays. What should you do?

Answer: C. Check the museum's official site before you rely on it

Facts in generated content need a source check.

- ✗ A. Trust it because the writing is polished and clear: polish is not evidence
- ✗ B. Assume all museums are free on Mondays: unsupported
- ✗ D. Ask the same tool to repeat the claim: repetition is not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F2-Q4 · question page 104

AI.BASICS.05 · Handle tasks that mix prediction and generation

A school wants AI to sort essays by topic and then write feedback. Which step should be checked first if the feedback seems off-topic?

Answer: A. The sorting step that labeled each essay's topic

A wrong label early in the chain can make later output off-topic.

- ✗ B. The length of each feedback paragraph: length is not the likely cause
- ✗ C. Whether essays should ever get written feedback: avoids the problem
- ✗ D. The paper the essays were printed on: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S5-F2-Q5 · question page 105

AI.BASICS.05 · Handle tasks that mix prediction and generation

A manager asks a generator, "Is this invoice fraudulent?" and gets a long, confident paragraph. What is the main concern?

Answer: D. Confident prose is not evidence the judgment is right

A decision needs evidence and testing, not just polished wording.

- ✗ A. Generators always refuse fraud questions: not true

✗ B. Long answers are always more accurate: length is not accuracy

✗ C. The long paragraph proves that the invoice is fraud: wording proves nothing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-M-Q1 · question page 107

AI.BASICS.06 · Tell a model from the product built around it

What is a model in an AI system?

Answer: C. The trained part that turns input into output

The model is the learned core; the product wraps it.

✗ A. The app's screen and buttons you tap: that is part of the product

✗ B. The company that sells the subscription: a company is not a model

✗ D. The list of usage limits on an account: limits are set by the product

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-M-Q2 · question page 108

AI.BASICS.06 · Tell a model from the product built around it

Two apps use the same model but behave differently. What is the most likely reason?

Answer: A. Each product adds its own instructions, tools and limits

Products wrap the same model in different ways.

✗ B. The model must be broken in one of the apps: differences usually come from the products

✗ C. Products cannot share a model, so this cannot happen: they can share models

✗ D. Models change randomly each time you open an app: not the usual reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S6-M-Q3 · question page 109

AI.BASICS.06 · Tell a model from the product built around it

A chatbot app can search the web. Where does that ability most likely come from?

Answer: B. A search tool the product connects to the model

Tools like web search are added by the product.

- ✗ A. Every model's built-in ability to browse: browsing is a product tool
- ✗ C. The user's own computer searching by itself: the product provides the tool
- ✗ D. The model memorizing the entire internet: a model is not a live copy of the web

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-M-Q4 · question page 110

AI.BASICS.06 · Tell a model from the product built around it

Your friend says, "Chatbots always remember everything you tell them." What is the best response?

Answer: D. It depends on the product; features differ

Memory is a product feature that varies.

- ✗ A. Only paid chatbots have models inside: all chatbots use some model
- ✗ B. That is true of every chatbot ever made: products differ
- ✗ C. No chatbot can store anything at all: some products save information

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-M-Q5 · question page 111

AI.BASICS.06 · Tell a model from the product built around it

Which is part of the product rather than the model?

Answer: C. A daily limit on how many messages you can send

Usage limits are set by the product.

- ✗ A. The patterns learned from example text: that is the model
- ✗ B. The learned patterns that produce replies: that is the model
- ✗ D. The numbers set during training: those belong to the model

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-B1-Q1 · question page 112

AI.BASICS.06 · Compare an engine with the car around it

In the comparison, what is the AI model like?

Answer: B. The engine

The model produces the output, like an engine.

- ✗ A. The steering wheel: part of the product
- ✗ C. The seats: part of the product
- ✗ D. The radio: part of the product

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-B1-Q2 · question page 113

AI.BASICS.06 · Compare an engine with the car around it

What is the product like?

Answer: A. The car around the engine

The product adds everything around the model.

- ✗ B. Only the fuel: not in the comparison
- ✗ C. Only the engine by itself: that is the model

✗ D. Only the road: not in the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-B1-Q3 · question page 114

AI.BASICS.06 · Compare an engine with the car around it

Why can two cars with the same engine drive differently?

Answer: D. Their other parts and settings differ

Same engine, different car.

✗ A. Engines change their shape every single day: not the reason
✗ B. Cars cannot share engines: they can
✗ C. One engine must be broken: not the usual reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-B1-Q4 · question page 115

AI.BASICS.06 · Compare an engine with the car around it

A van is slow because of a speed limiter. In AI terms, the slowness comes from what?

Answer: C. The product

A limit set by the product, not the model.

✗ A. The user: not the cause here
✗ B. The model: the engine is not the cause here
✗ D. The training data: not the cause here

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-B1-Q5 · question page 116

AI.BASICS.06 · Compare an engine with the car around it

Which is like a product feature added around the model?

Answer: B. A safety filter

Filters are added by the product.

- ✗ A. The trained numbers: model
- ✗ C. The model's output: produced by the model
- ✗ D. The learned patterns: model

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F1-Q1 · question page 117

AI.BASICS.06 · Apply the model and product distinction

A school's writing app refuses to write full essays, while another app using the same model will. What explains this?

Answer: A. The school's product added its own rules

Products can add rules on top of the same model.

- ✗ B. The model in the school app is too weak: the model is the same
- ✗ C. The other app has a different kind of model: the model is the same
- ✗ D. Essay writing is impossible for any model: the other app does it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F1-Q2 · question page 118

AI.BASICS.06 · Apply the model and product distinction

A coworker says, "The model can read our team calendar." What is more accurate?

Answer: D. The product may connect a calendar tool to the model

Access to tools like calendars comes from the product.

- ✗ A. Calendars are part of every model's patterns: they are not

✗ B. Models learn calendars during training: your calendar was not in training

✗ C. Every model can read every calendar automatically: access comes from product tools

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F1-Q3 · question page 119

AI.BASICS.06 · Apply the model and product distinction

Ana read an article about one chatbot's limits. What should she remember about a different chatbot?

Answer: C. Its limits may differ, so check that product

Product details do not transfer automatically.

✗ A. It uses no model if it is different: every chatbot uses a model

✗ B. It has no limits since it is a different brand: unsupported

✗ D. Its limits must match the article exactly: products differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F1-Q4 · question page 120

AI.BASICS.06 · Apply the model and product distinction

A product updates to a new model but keeps its name. What could change for users?

Answer: D. Its answers and abilities could change

Swapping the model can change behavior under the same name.

✗ A. Only the price, never the answers: answers can change too

✗ B. Only the colors of the app: the model affects output

✗ C. Nothing, because the name stayed the same: the model changed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F1-Q5 · question page 121

AI.BASICS.06 · Apply the model and product distinction

Which statement about chatbots is accurate?

Answer: B. Different chatbots can answer one question differently

Products and models vary, so behavior varies.

- ✗ A. Every chatbot can browse the web and save files: tools vary by product
- ✗ C. All chatbots give the same answer to any question: products and models differ
- ✗ D. A chatbot is a model with no product around it: chatbots are products

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F2-Q1 · question page 122

AI.BASICS.06 · Check current product details instead of assuming

A blog post from two years ago says a product cannot read PDF files. You need to know today. What should you check?

Answer: A. The product's current help pages or release notes

Check current, product-specific information.

- ✗ B. The same two-year-old blog post again: it may be outdated
- ✗ C. Nothing; products never change their features: they do change
- ✗ D. A friend's experience with another product: a different product

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F2-Q2 · question page 123

AI.BASICS.06 · Check current product details instead of assuming

A photo tagger and a chatbot are both called "AI." Can you use the photo tagger to draft a letter?

Answer: D. No; they are different products for different tasks

Different products serve different tasks.

- ✗ A. Yes; all AI products can do every task: tools are not interchangeable
- ✗ B. Yes; if both use AI, they must share one model: they may use different models
- ✗ C. No; letters can never be drafted with AI: chatbots can draft letters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F2-Q3 · question page 124

AI.BASICS.06 · Check current product details instead of assuming

Your settings page says "Chat history saving: off," but a friend says the product always saves chats. What is the best next step?

Answer: B. Read what the product says the setting does and rely on that

Product documentation for your own settings is the best evidence.

- ✗ A. Delete your account without checking anything: check first
- ✗ C. Assume your friend is right and ignore the setting entirely: hearsay is weaker evidence
- ✗ D. Assume no product ever saves anything: some do

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F2-Q4 · question page 125

AI.BASICS.06 · Check current product details instead of assuming

A product changed its limits last month. Your guide from last year says "unlimited messages." What should you conclude?

Answer: C. The guide may be out of date; check the current terms

Current product information beats old guides.

✗ A. All guides are always wrong about limits: too broad

✗ B. Limits are set by the model, so they never change: limits are set by the product

✗ D. The guide must be right because it was published: products change

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S6-F2-Q5 · question page 126

AI.BASICS.06 · Check current product details instead of assuming

A company says its product "uses the latest model." Why might that be hard to rely on later?

Answer: A. The product may switch models over time

A model can be swapped behind the same product name.

✗ B. Models never change once released: they are updated

✗ C. "Latest" is a guarantee of accuracy: it is not

✗ D. Products cannot contain models: they do

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-M-Q1 · question page 128

AI.BASICS.07 · Use human comparisons carefully

A system scores better than doctors at spotting one type of spot on X-rays. Which conclusion is justified?

Answer: D. It did well on that narrow task; other tasks need checks

Narrow success does not transfer automatically.

✗ A. It will also be excellent at reading patient notes: nearby tasks can differ

✗ B. It can now replace doctors for all medical decisions: one task does not predict others

✗ C. It must understand medicine as well as a doctor: performance is not understanding

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-M-Q2 · question page 129

AI.BASICS.07 · Use human comparisons carefully

Why can AI abilities be called "uneven"?

Answer: B. It may do a hard task well but fail an easy one

Abilities can be strong and weak in surprising combinations.

- ✗ A. It improves evenly like a growing child: abilities do not grow evenly
- ✗ C. It always performs the same on every task: that is even, not uneven
- ✗ D. It only fails at tasks that are very hard: it can fail easy tasks

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-M-Q3 · question page 130

AI.BASICS.07 · Use human comparisons carefully

A teacher says, "The AI is like a student learning to write." What is the best way to use this comparison?

Answer: A. As a starting point, then check how it actually performs

Comparisons help explain but must be checked.

- ✗ B. As a reason never to check its writing: checking is still needed
- ✗ C. As proof that the AI learns exactly like a student: comparisons are not proof
- ✗ D. As a reason to give it grades like a student: misreads the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-M-Q4 · question page 131

AI.BASICS.07 · Use human comparisons carefully

A system writes fluent essays but miscounts the apples in a simple picture. What does this show?

Answer: C. Skill at one task does not predict skill at another

Uneven abilities are common.

- ✗ A. Counting is always harder than writing for computers: unsupported generalization
- ✗ B. The essays must also be full of counting errors: not necessarily
- ✗ D. It is pretending to be bad at counting on purpose: no evidence of pretending

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-M-Q5 · question page 132

AI.BASICS.07 · Use human comparisons carefully

Which statement is a careful description rather than a misleading comparison?

Answer: B. "It recognized 95 of 100 test images correctly."

A measured result on a task can be checked.

- ✗ A. "It sees the world in just the way we do.": a misleading comparison
- ✗ C. "It is as smart as a ten-year-old.": a misleading overall comparison
- ✗ D. "It thinks like a genius.": vague and unsupported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-B1-Q1 · question page 133

AI.BASICS.07 · Compare uneven skill to a calculator and a toddler

What can the calculator do better than a person?

Answer: C. Multiply huge numbers quickly

Calculators excel at arithmetic.

✗ A. Tell a cat from a dog at a glance: it cannot

✗ B. Explain a joke: not described

✗ D. Learn a language: not described

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-B1-Q2 · question page 134

AI.BASICS.07 · Compare uneven skill to a calculator and a toddler

What can the toddler do that the calculator cannot?

Answer: D. Tell a cat from a dog

The toddler recognizes animals.

✗ A. Add huge numbers: the calculator does that

✗ B. Run on batteries: not the point

✗ C. Multiply 27 by 43: the toddler cannot

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-B1-Q3 · question page 135

AI.BASICS.07 · Compare uneven skill to a calculator and a toddler

Which question does the passage recommend?

Answer: A. "How good is it at this task?"

Ask about a specific task.

✗ B. "Does it think like us?": not checkable

✗ C. "How smart is it overall?": too vague

✗ D. "Is it smarter than a toddler?": misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-B1-Q4 · question page 136

AI.BASICS.07 · Compare uneven skill to a calculator and a toddler

Who is "smarter" overall: the calculator or the toddler?

Answer: B. Neither; they have different abilities

Abilities differ; one score misleads.

- ✗ A. Both are equally smart at everything: not true
- ✗ C. The calculator, since it is faster: speed at one task is not overall smartness
- ✗ D. The toddler, since it is human: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-B1-Q5 · question page 137

AI.BASICS.07 · Compare uneven skill to a calculator and a toddler

What does the example suggest about AI?

Answer: A. It can be strong at one thing and weak at another

Uneven abilities apply to AI too.

- ✗ B. It is always stronger than a calculator at math: not true
- ✗ C. It is equally good at all tasks: abilities are uneven
- ✗ D. It is always weaker than a toddler: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F1-Q1 · question page 138

AI.BASICS.07 · Judge AI ability task by task

A tool translated a menu well. You now want it to translate a medical form. What should you do?

Answer: D. Check its performance on the medical form separately

Check each task that matters.

✗ A. Assume medical words are easier to translate than menus: unsupported

✗ B. Refuse because translation is impossible: it can translate

✗ C. Trust it fully since the menu was perfect: one task does not predict another

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F1-Q2 · question page 139

AI.BASICS.07 · Judge AI ability task by task

Jordan says, "This chatbot is basically a person in a box." What is the most accurate reply?

Answer: C. It can sound human, but its abilities differ from a person's

Avoid both over- and under-estimating.

✗ A. Yes, it has the same judgment and common sense as a person: unsupported

✗ B. No, it can never do anything useful: dismissive and inaccurate

✗ D. Yes, because it uses words like a person: sounding human is not being human

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F1-Q3 · question page 140

AI.BASICS.07 · Judge AI ability task by task

A study tool explains algebra steps clearly but gets one multiplication wrong. What is the best habit?

Answer: B. Use the explanation but check each calculation

Uneven ability means checking specific parts.

✗ A. Assume all its explanations are wrong: too broad

✗ C. Trust every step because the explanation is clear: clarity is not correctness

✗ D. Stop using it for anything at all: too extreme

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F1-Q4 · question page 141

AI.BASICS.07 · Judge AI ability task by task

Which report about an AI tool is most useful to a manager?

Answer: C. "It sorted 180 of 200 invoices; handwritten ones failed."

A specific measured observation helps decisions.

✗ A. "It is dumb and will never work.": vague and dismissive
✗ B. "It thinks just like our accountant.": misleading comparison
✗ D. "It is really smart, just like a new hire we trained.": vague comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F1-Q5 · question page 142

AI.BASICS.07 · Judge AI ability task by task

An image tool wins an art contest. A friend says it can surely design a safe bridge. What is wrong?

Answer: D. Success in one task does not show skill in a different one

Abilities do not transfer automatically.

✗ A. Art and bridge design are the same task: they differ greatly
✗ B. Image tools cannot win art contests: irrelevant to the reasoning
✗ C. Nothing; winning a contest proves its general skill: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F2-Q1 · question page 143

AI.BASICS.07 · Describe what was observed instead of "smart" or "dumb"

A chatbot answers warmly, "I totally understand how you feel." What should you keep in mind?

Answer: A. Friendly wording does not show feelings or judgment

Sounding human can raise trust without evidence.

- ✗ B. It is lying, so nothing it says is useful: too extreme
- ✗ C. Friendly replies are always more accurate than others: tone is not accuracy
- ✗ D. It must truly share your feelings: unsupported by the wording

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F2-Q2 · question page 144

AI.BASICS.07 · Describe what was observed instead of "smart" or "dumb"

Which statement is a checkable observation?

Answer: D. "It summarized ten reports but invented one date."

Specific observations can be verified.

- ✗ A. "It understands us better than other people do.": not checkable
- ✗ B. "It is just a dumb machine.": not checkable
- ✗ C. "It is basically genius-level smart.": not checkable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F2-Q3 · question page 145

AI.BASICS.07 · Describe what was observed instead of "smart" or "dumb"

A coworker dismisses a tool as "just a machine" and ignores that it can write convincing fake emails. What is the risk?

Answer: C. Underestimating real abilities can hide real risks

Dismissing a system can be as misleading as overpraising it.

✗ A. Calling it a machine makes it safer: labels do not change risk

✗ B. None, since machines cannot write convincing emails: it can

✗ D. The tool will stop working if ignored: not how tools work

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F2-Q4 · question page 146

AI.BASICS.07 · Describe what was observed instead of "smart" or "dumb"

Why might a friendly, human-sounding style make errors easier to miss?

Answer: A. People tend to trust things that sound like a helper

Tone affects trust without changing accuracy.

✗ B. Errors are always marked in red: not true

✗ C. Human-sounding systems cannot make real mistakes: they can

✗ D. Friendly text contains no errors by design: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S7-F2-Q5 · question page 147

AI.BASICS.07 · Describe what was observed instead of "smart" or "dumb"

A review says, "The AI is honest and caring." What would make the review more useful?

Answer: B. Describe specific tasks it did well or badly

Specific observations can be checked; personality labels cannot.

✗ A. Say it is "better than a human": unsupported overall claim

✗ C. Compare it to a famous person: still a vague comparison

✗ D. Use stronger words like "loving" and "wise": still not checkable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-M-Q1 · question page 149

AI.LEARN.01 · Tell training from inference (using the model)

A company shows a model millions of labeled photos and adjusts it until it sorts them well. Which stage is this?

Answer: D. Training

Adjusting the model to match examples is training.

- ✗ A. Inference, because the photos are labeled: labels are used for learning
- ✗ B. Neither, because sorting photos is not AI: sorting photos is a common AI task
- ✗ C. Inference: it is still learning from examples

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-M-Q2 · question page 150

AI.LEARN.01 · Tell training from inference (using the model)

You type a question into a finished chatbot and get a reply. Which stage is this?

Answer: C. Inference: the finished model handles new input

Using a finished model on new input is inference.

- ✗ A. Training: the model rewrites itself for your question: its learned settings do not change
- ✗ B. Neither stage, because no model is involved: a model produces the reply
- ✗ D. Training: your question becomes part of its settings: ordinary use does not change the settings

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-M-Q3 · question page 151

AI.LEARN.01 · Tell training from inference (using the model)

Which statement about training is accurate?

Answer: A. It adjusts the model's settings using many examples

Training means repeated adjustment using many examples.

- ✗ B. It happens each time anyone opens the app on a phone: training happens before use
- ✗ C. It is the same as typing a question to the model: that is inference
- ✗ D. It needs only one example to finish completely: training uses large amounts of data

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-M-Q4 · question page 152

AI.LEARN.01 · Tell training from inference (using the model)

During ordinary use (inference), what happens to the model's learned settings?

Answer: B. They stay the same

Inference uses the settings; it does not change them.

- ✗ A. They reset to random values: they keep what was learned
- ✗ C. They update after every message: ordinary use does not update them
- ✗ D. They copy the user's whole device: not what happens

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-M-Q5 · question page 153

AI.LEARN.01 · Tell training from inference (using the model)

Why does training usually cost much more computing power than a single use?

Answer: C. It repeats adjustments over huge amounts of data

Training processes vast data many times.

- ✗ A. It must print out every training example on paper: not how training works
- ✗ B. It only ever handles one short question at a time: that describes a single

use

✗ D. It happens on the user's phone after each reply: training is usually done in advance

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-B1-Q1 · question page 154

AI.LEARN.01 · Compare studying for a driving test with driving

Practicing with a driving teacher is like which stage?

Answer: D. Training

Practice and correction are training.

✗ A. Inference: practice is the learning stage
✗ B. Neither stage: it matches training
✗ C. Both stages equally: practice is learning, not use

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-B1-Q2 · question page 155

AI.LEARN.01 · Compare studying for a driving test with driving

Driving to the store with your license is like which stage?

Answer: A. Inference

Using skills on a new road is inference.

✗ B. Neither stage: it matches inference
✗ C. Training: you are using what you learned
✗ D. A driving exam: not the comparison made

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-B1-Q3 · question page 156

AI.LEARN.01 · Compare studying for a driving test with driving

Which stage usually happens first?

Answer: B. Training

Learning comes before use.

- ✗ A. Neither happens: both happen
- ✗ C. They happen at once: learning comes first
- ✗ D. Inference: you use skills after learning them

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-B1-Q4 · question page 157

AI.LEARN.01 · Compare studying for a driving test with driving

On the trip to the store, are you taking a lesson?

Answer: A. No, you are using what you learned

Using a skill is different from learning it.

- ✗ B. Yes, every trip is a full lesson: the trip is not a lesson
- ✗ C. No, because you forgot how to drive: you remember your skills
- ✗ D. Yes, the store teaches you to drive: the store is not a teacher

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-B1-Q5 · question page 158

AI.LEARN.01 · Compare studying for a driving test with driving

Which stage takes a long time and many corrections?

Answer: C. Training

Learning takes time and correction.

- ✗ A. Driving to the store: that is use

✗ B. Inference: using is quick

✗ D. Getting the license card: not the stage described

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F1-Q1 · question page 159

AI.LEARN.01 · Identify training and inference in real situations

A hospital's team spends weeks adjusting a model with thousands of past X-rays. Later, a doctor uploads a new X-ray for a second opinion. Which part is inference?

Answer: D. The doctor uploading the new X-ray

Running the finished model on a new X-ray is inference.

✗ A. Hiring the team that adjusted the model: not a model stage

✗ B. Collecting the thousands of past X-rays: that prepares training data

✗ C. The weeks of adjusting the model: that is training

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F1-Q2 · question page 160

AI.LEARN.01 · Identify training and inference in real situations

Mia asks a translation app to translate a menu. Does her request train the app's model?

Answer: B. No; translating her menu is inference

Using the app is inference.

✗ A. Yes; every request is a training example instantly: ordinary use does not change settings

✗ C. No; translation apps never use any models: they do use models

✗ D. Yes; the model rewrites its settings for menus: not during ordinary use

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F1-Q3 · question page 161

AI.LEARN.01 · Identify training and inference in real situations

A company says its new model "finished training in June." What does that tell you?

Answer: D. Learning from its data ended then; later use is inference

Finishing training means the model's settings were fixed then.

- ✗ A. It cannot answer any questions after June: it can be used after training
- ✗ B. It was trained by answering users' questions in June: not implied
- ✗ C. It keeps training on every question users type: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F1-Q4 · question page 162

AI.LEARN.01 · Identify training and inference in real situations

Which activity is training?

Answer: B. Adjusting a model's settings with labeled examples

Changing settings with examples is training.

- ✗ A. Asking a finished model for a recipe idea: inference
- ✗ C. Reading the reply a model already produced: not a model stage
- ✗ D. Uploading a photo to be tagged by an app: inference

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F1-Q5 · question page 163

AI.LEARN.01 · Identify training and inference in real situations

Why is it useful to know that ordinary use does not change a model's settings?

Answer: A. Correcting it in one chat will not fix the model itself

A correction in a chat affects that conversation, not the trained model.

✗ B. It means the model gets smarter with each question you ask: the opposite of what was taught

✗ C. It means the model can never be updated by its makers: makers can train new versions

✗ D. It means the model forgets everything after each reply: it keeps its learned settings

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F2-Q1 · question page 164

AI.LEARN.01 · Ask what actually changes when a product seems to "learn"

You click "thumbs down" on a wrong answer. What is the most accurate description?

Answer: C. It is feedback the company may use later, depending on its policy

Feedback may inform later training, depending on policy.

✗ A. It instantly retrains the model to fix that answer for everyone: feedback does not retrain instantly

✗ B. It deletes the model and replaces it with a better one at once: not what feedback does

✗ D. It has no possible use for anyone at all: companies may use feedback

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F2-Q2 · question page 165

AI.LEARN.01 · Ask what actually changes when a product seems to "learn"

A chatbot remembers your name in a new conversation. What is the most likely explanation?

Answer: B. The product stored your name and fed it back in

Saved information is a product feature, not a change to the model.

✗ A. The model guessed your name from its training data: it was not in training data

✗ C. Every model is born knowing every user's name: not true

✗ D. The model retrained itself on your name overnight: ordinary use does not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F2-Q3 · question page 166

AI.LEARN.01 · Ask what actually changes when a product seems to "learn"

Sam wants to know whether his uploaded files could be used to train future models. What should he check?

Answer: C. The provider's data policy and his account settings

Data use for training is set by policy and settings.

✗ A. Whether the chatbot sounds friendly when he asks: tone is not policy

✗ B. Whether the chatbot remembers his name: memory is not a training policy

✗ D. Nothing; uploaded files are never used by anyone: depends on the provider

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F2-Q4 · question page 167

AI.LEARN.01 · Ask what actually changes when a product seems to "learn"

A user corrects a chatbot's wrong date and it apologizes. Later, a different user asks the same question. What should we expect?

Answer: D. The model may make the same mistake for the other user

A correction in one conversation does not change the trained model.

✗ A. The model will now always give the correct date: one chat does not retrain the model

✗ B. The model will refuse to answer that question again: no reason to expect this

✗ C. The other user will see the first user's correction: conversations are separate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S8-F2-Q5 · question page 168

AI.LEARN.01 · Ask what actually changes when a product seems to "learn"

Which change would actually alter the model's learned settings?

Answer: A. The company training a new version with new data

Only training changes the model's settings.

✗ B. A user rating one reply with a thumbs-up: feedback alone does not retrain

✗ C. A user typing "remember this" in a chat: that may store a note, not retrain

✗ D. A user starting a brand-new conversation: that changes the context, not the model

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-M-Q1 · question page 170

AI.LEARN.02 · Describe parameters as learned settings that capture patterns

What are parameters in a model?

Answer: D. The adjustable numbers set during training

Parameters are numbers the training process adjusts.

✗ A. The questions users type into the finished app: those are inputs

✗ B. The list of every training example, kept word for word: the model keeps settings, not a full copy

✗ C. The rules a programmer wrote out line by line: parameters are learned, not written

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-M-Q2 · question page 171

AI.LEARN.02 · Describe parameters as learned settings that capture patterns

What happens to the parameters when the model gets a training example wrong?

Answer: A. They are nudged so the next output is closer

Small adjustments add up over many examples.

- ✗ B. They are replaced with the example itself: examples are not stored that way
- ✗ C. They are erased and started over from zero: they are adjusted a little
- ✗ D. They stay exactly the same as before: training changes them

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-M-Q3 · question page 172

AI.LEARN.02 · Describe parameters as learned settings that capture patterns

After training, what does a model mainly keep?

Answer: B. Dial settings that capture patterns in the data

Learned patterns live in the parameter settings.

- ✗ A. A tidy, searchable copy of every example it saw: it keeps settings, not a full copy
- ✗ C. Nothing at all; it relearns each time it is used: it keeps its settings
- ✗ D. Only the last example it saw during training: it keeps patterns from all of training

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-M-Q4 · question page 173

AI.LEARN.02 · Describe parameters as learned settings that capture patterns

A model has twice as many parameters as another. Which statement is accurate?

Answer: C. It can capture more patterns, but bigger is not always better

More parameters add capacity, not a guarantee.

- ✗ A. It cannot be used by ordinary people at all: unrelated
- ✗ B. It must have been trained on exactly half the data: no such rule
- ✗ D. It is guaranteed to be twice as accurate on every task: size alone guarantees nothing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-M-Q5 · question page 174

AI.LEARN.02 · Describe parameters as learned settings that capture patterns

Can a model ever reproduce a piece of text it saw during training?

Answer: B. Sometimes, especially text it saw many times

Frequently repeated text can be reproduced.

- ✗ A. Always, because it stores every page exactly: it does not store a full copy
- ✗ C. Only if the user types the password for it: no such rule
- ✗ D. Never, because it keeps no trace of any text: strong traces can remain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-B1-Q1 · question page 175

AI.LEARN.02 · Picture tuning a guitar by ear

In the comparison, what are the guitar pegs like?

Answer: A. The model's parameters

Pegs are adjustable settings, like parameters.

✗ B. The app's screen: not in the comparison

✗ C. The training examples: the plucked notes are more like examples

✗ D. The user's questions: not in the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-B1-Q2 · question page 176

AI.LEARN.02 · Picture tuning a guitar by ear

What do you do when a string sounds flat?

Answer: D. Turn the peg a little and try again

Small corrections, repeated, are how tuning works.

✗ A. Stop playing for good: you adjust and retry

✗ B. Throw the guitar away: not how tuning works

✗ C. Write down the sound and keep it forever: you keep the setting, not the sound

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-B1-Q3 · question page 177

AI.LEARN.02 · Picture tuning a guitar by ear

After tuning, what do you keep?

Answer: C. The final peg positions

You keep settings, not every sound.

✗ A. The broken strings: not in the passage

✗ B. A recording of every note plucked: not kept

✗ D. Nothing at all: the positions stay set

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-B1-Q4 · question page 178

AI.LEARN.02 · Picture tuning a guitar by ear

Which part of training is like plucking a string and hearing it is wrong?

Answer: A. Seeing that an output does not match an example

Checking output against examples drives adjustment.

- ✗ B. Deleting the model to start over: not the comparison
- ✗ C. Buying a faster computer for the model: not the comparison
- ✗ D. Typing a new question into the finished app: that is use, not training

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-B1-Q5 · question page 179

AI.LEARN.02 · Picture tuning a guitar by ear

Why are there "millions of pegs" in a model?

Answer: B. It has very many parameters

Models have huge numbers of parameters.

- ✗ A. It has many users: not the point
- ✗ C. It runs on many computers: not the point
- ✗ D. It plays very many songs: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F1-Q1 · question page 180

AI.LEARN.02 · Apply the idea of parameters to claims about models

A news article says a model "contains every book it read." What is more accurate?

Answer: D. It holds learned settings, not a complete copy of every book

Parameters capture patterns, not full copies, though fragments can be

reproduced.

- ✗ A. Models never learn anything from books at all: text is common training data
- ✗ B. It is accurate; parameters are the books themselves: parameters are numbers
- ✗ C. It is accurate; models are libraries of full texts: models keep settings, not full copies

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F1-Q2 · question page 181

AI.LEARN.02 · Apply the idea of parameters to claims about models

Two models: one has 1 billion parameters, the other 10 billion. Which conclusion is justified without testing?

Answer: C. The larger has more capacity; testing shows which works better

Capacity is not the same as tested performance.

- ✗ A. Both must give identical answers: not justified
- ✗ B. The smaller one must be useless for every task: not justified
- ✗ D. The larger one is exactly ten times more accurate on every task: size does not set accuracy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F1-Q3 · question page 182

AI.LEARN.02 · Apply the idea of parameters to claims about models

A model was trained to recognize cats. What did training adjust?

Answer: B. Its parameters, so pixel patterns map to "cat"

Training set the parameters to capture cat patterns.

- ✗ A. The photos themselves, so they look more like cats: training changes

the model, not photos

✗ C. A written rule list describing every cat breed: it learned rather than followed rules

✗ D. The user's camera settings: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F1-Q4 · question page 183

AI.LEARN.02 · Apply the idea of parameters to claims about models

Why might a model repeat a famous quotation word for word?

Answer: D. Text seen many times in training can leave strong traces

Frequent repetition in training can cause memorization.

✗ A. It looks the quotation up in a stored copy of the internet: a model is not a stored copy

✗ B. Models are designed to repeat everything word for word: they usually do not

✗ C. The quotation is one of its parameters: parameters are numbers

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F1-Q5 · question page 184

AI.LEARN.02 · Apply the idea of parameters to claims about models

Which description of training is most accurate?

Answer: C. Many small adjustments to settings, based on errors

Training repeats small, error-driven nudges.

✗ A. One big change made after all examples are seen: it adjusts many times

✗ B. Typing in facts one at a time into a database: not how training works

✗ D. Copying the examples into the model's memory: it learns patterns instead

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F2-Q1 · question page 185

AI.LEARN.02 · Avoid oversimplified claims about what a model stores

A user asks a company to "delete the fact about me from the model." Why is this hard?

Answer: A. Learned patterns are spread across settings, not stored as one fact

Influence is spread across many parameters.

- ✗ B. Models store each fact in a labeled folder that is locked: not how models store patterns
- ✗ C. It is easy; any user can delete one parameter: not practical
- ✗ D. Models never learn anything about people: they can learn from text about people

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F2-Q2 · question page 186

AI.LEARN.02 · Avoid oversimplified claims about what a model stores

A model's answer blends ideas so that no single source said exactly that. What explains this?

Answer: D. It combines learned patterns from many examples

Blending patterns can create statements no source made.

- ✗ A. It always copies one source exactly: it often blends
- ✗ B. It searched one website and pasted it: not implied
- ✗ C. It has a written rule never to use any real sources: no such rule

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F2-Q3 · question page 187

AI.LEARN.02 · Avoid oversimplified claims about what a model stores

Someone says, "Models never copy anything, so there is no copyright issue." What is wrong?

Answer: B. Models can sometimes reproduce passages they saw many times

"Never copies" is too simple.

- ✗ A. Nothing; models are proven never to copy at all: they sometimes reproduce text
- ✗ C. Models only copy images, never text: not accurate
- ✗ D. Copying is impossible because parameters are numbers: numbers can still encode strong traces

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F2-Q4 · question page 188

AI.LEARN.02 · Avoid oversimplified claims about what a model stores

Someone says, "Models just copy and paste their training data." What is wrong?

Answer: A. Mostly they produce new combinations from patterns

"Just copies" is also too simple.

- ✗ B. Models only copy text from the user: not accurate
- ✗ C. Models cannot produce any text at all: they can
- ✗ D. Nothing; every output is an exact copy of a source: most outputs are new combinations

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S9-F2-Q5 · question page 189

AI.LEARN.02 · Avoid oversimplified claims about what a model stores

Why can't you open a model and read "the fact" it learned about the moon?

Answer: C. It is spread across many numbers, not stored as sentences

Parameters are numbers, not readable sentences.

✗ A. Models never learn about the moon: they can learn patterns about it

✗ B. The fact is locked away inside it with a secret password: not the reason

✗ D. Models refuse to discuss the moon: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-M-Q1 · question page 191

AI.LEARN.03 · Explain training, validation and test data

What is the training set used for?

Answer: D. The examples the model learns from

The model learns from the training set.

✗ A. Advertising the model to customers: not a data split

✗ B. Comparing versions during development: that is the validation set

✗ C. The final check: that is the test set

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-M-Q2 · question page 192

AI.LEARN.03 · Explain training, validation and test data

What is the test set used for?

Answer: C. Measuring the finished model on unseen examples

The test set gives a final, honest measurement.

✗ A. Tuning settings during development: that is validation

✗ B. Teaching the model its main patterns: that is training

✗ D. Storing users' questions after the app launches: not a data split

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-M-Q3 · question page 193

AI.LEARN.03 · Explain training, validation and test data

Which set is most like practice quizzes during development?

Answer: B. The validation set

Validation helps choose versions and settings.

- ✗ A. None of the sets: validation matches practice quizzes
- ✗ C. The test set: the test is the final exam
- ✗ D. The training set: that is the study material

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-M-Q4 · question page 194

AI.LEARN.03 · Explain training, validation and test data

Why must test examples stay out of training?

Answer: A. Otherwise the test score can overstate real performance

Seen examples inflate scores.

- ✗ B. Otherwise the training will finish too quickly: not the reason
- ✗ C. Otherwise the model will refuse to work on any input at all: not the reason
- ✗ D. Otherwise the model gets too many parameters: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-M-Q5 · question page 195

AI.LEARN.03 · Explain training, validation and test data

A model scores 99 percent, but later it is found that test examples were also in training. What should you conclude?

Answer: B. The 99 percent may overstate how it handles new examples

Leakage makes a score look better than real performance.

✗ A. Nothing; leaks never affect scores: leaks can inflate scores

✗ C. The model must be useless for everything: it may still work, but the score misleads

✗ D. The 99 percent is now even more trustworthy than before: leakage makes it less trustworthy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-B1-Q1 · question page 196

AI.LEARN.03 · Compare studying, practice quizzes and a final exam

The textbook is like which data set?

Answer: A. The training set

The model learns from the training set.

✗ B. The validation set: those are practice quizzes

✗ C. The test set: that is the final exam

✗ D. No data set: it matches training

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-B1-Q2 · question page 197

AI.LEARN.03 · Compare studying, practice quizzes and a final exam

The final exam is like which data set?

Answer: D. The test set

The test set is the final check.

✗ A. The training set: that is the textbook

✗ B. All three sets: it matches the test set

✗ C. The validation set: those are quizzes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-B1-Q3 · question page 198

AI.LEARN.03 · Compare studying, practice quizzes and a final exam

Why must the final exam questions stay secret?

Answer: C. So a high score shows understanding

Unseen questions give an honest measure.

- ✗ A. So the textbook is not needed: not the reason
- ✗ B. So the exam is harder to print: not the reason
- ✗ D. So the teacher can grade all the exams faster: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-B1-Q4 · question page 199

AI.LEARN.03 · Compare studying, practice quizzes and a final exam

What do practice quizzes help the student do?

Answer: D. See what still needs work

Quizzes guide preparation.

- ✗ A. Get the final grade early: they are practice
- ✗ B. Skip the textbook: not their purpose
- ✗ C. Memorize the final exam questions early: the exam stays secret

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-B1-Q5 · question page 200

AI.LEARN.03 · Compare studying, practice quizzes and a final exam

A student saw the exact exam in advance and scored 100. What does the score mostly show?

Answer: B. Memory of those questions

Seen questions measure memory.

✗ A. That the exam was too easy: not the main issue

✗ C. Deep understanding of everything: not shown

✗ D. Nothing at all: it shows memory

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F1-Q1 · question page 201

AI.LEARN.03 · Apply data splits to real projects

A team building a dog-breed recognizer has 10,000 labeled photos. Which plan follows good practice?

Answer: C. Train on most, use some for validation, keep a test portion unseen

Separate splits give an honest final measure.

✗ A. Train and test on the same 10,000 photos: testing on training data inflates scores

✗ B. Test first, then add the test photos into training and report that score: the reported score would be inflated

✗ D. Skip testing because 10,000 is already enough: testing is still needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F1-Q2 · question page 202

AI.LEARN.03 · Apply data splits to real projects

During development, a team compares three versions and picks the best. Which data should they use to compare?

Answer: A. The validation set

Validation data is for comparing versions.

✗ B. The training set only: scores there are too optimistic

✗ C. No data; they should guess: data is needed

✗ D. The test set, again and again: the test set is for the final check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F1-Q3 · question page 203

AI.LEARN.03 · Apply data splits to real projects

A student project reports "100 percent accuracy," measured on the same examples it trained on. What is the problem?

Answer: B. The score does not show how it handles new examples

Only unseen examples show real performance.

- ✗ A. There is no problem; 100 percent is always trustworthy: it was measured on seen data
- ✗ C. Training data cannot be used for anything at all: it is used to learn
- ✗ D. Accuracy cannot be measured for student projects: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F1-Q4 · question page 204

AI.LEARN.03 · Apply data splits to real projects

When should the test set be used?

Answer: C. At the end, to measure the finished model

The test set is a final, one-time-style check.

- ✗ A. At the start, to teach the model: that is training
- ✗ B. Every day, to choose settings: that wears it out; use validation
- ✗ D. Never, because final tests are always optional: testing is essential

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F1-Q5 · question page 205

AI.LEARN.03 · Apply data splits to real projects

A company says its spam filter is "98 percent accurate on our test set." What question checks whether the number is trustworthy?

Answer: A. Were the test emails kept out of training?

An honest test score needs unseen test data.

- ✗ B. Is 98 a bigger number than last year's?: does not check leakage
- ✗ C. Does the filter have a friendly name?: irrelevant
- ✗ D. Is the number written in a large font?: presentation is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F2-Q1 · question page 206

AI.LEARN.03 · Spot hidden problems with test data

The same news story appears on two websites. One copy is in training, the other in the test set. What is the risk?

Answer: D. The test includes something the model has already seen

Near-duplicates can leak test content into training.

- ✗ A. The model will refuse to read news: not the risk
- ✗ B. The test becomes harder than real use: it becomes easier, not harder
- ✗ C. None, because the websites have different names: the content is the same

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F2-Q2 · question page 207

AI.LEARN.03 · Spot hidden problems with test data

A team checks its test score after every small change for months. What problem can this cause?

Answer: A. The team slowly tunes the model to that one test

Overusing the test set makes it less honest.

✗ B. The model forgets its training data: not the problem

✗ C. Nothing; checking often always helps: reusing the test set wears it out

✗ D. The test questions physically wear out from use: not literally

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F2-Q3 · question page 208

AI.LEARN.03 · Spot hidden problems with test data

A plant-identification app was tested on clear studio photos, but users upload blurry phone pictures. What should the team conclude?

Answer: C. The test score may not predict real-world performance

A test set should resemble real use.

✗ A. The test score proves the app works for everyone: the test did not match real use

✗ B. Blurry photos cannot be identified by anyone: too extreme

✗ D. The app will work even better on blurry photos: unsupported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F2-Q4 · question page 209

AI.LEARN.03 · Spot hidden problems with test data

Photos of the same dog, cropped slightly differently, end up in both training and test sets. What should the team do?

Answer: B. Remove near-duplicates so test items are truly unseen

Near-duplicates should be kept on one side only.

✗ A. Keep them, because cropping makes them different: they are near-copies

✗ C. Add more copies to make the test fairer: adds more leakage

✗ D. Delete the whole test set and skip testing: testing is still needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S10-F2-Q5 · question page 210

AI.LEARN.03 · Spot hidden problems with test data

Which test plan best predicts real performance for a store's receipt scanner?

Answer: D. Unseen receipts like the ones customers really scan

Unseen, realistic test data gives a useful score.

- ✗ A. Receipts the scanner was trained on: seen data
- ✗ B. One receipt scanned 100 times: too narrow
- ✗ C. Perfect, printed sample receipts from the factory only: not like real use

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-M-Q1 · question page 212

AI.LEARN.04 · Recognize overfitting and good generalization

What does it mean for a model to generalize well?

Answer: B. It does well on new examples, not just training ones

Generalizing means handling new examples well.

- ✗ A. It memorizes every one of its training examples exactly: that can be overfitting
- ✗ C. It works only on the examples it trained on: that is the opposite
- ✗ D. It has as many parameters as possible: size is not generalization

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-M-Q2 · question page 213

AI.LEARN.04 · Recognize overfitting and good generalization

Which result is a warning sign of overfitting?

Answer: C. 99 percent on training data but 70 percent on test data

A large training-to-test gap suggests overfitting.

- ✗ A. A model that has not been trained yet: not overfitting
- ✗ B. 50 percent on training data and 50 percent on test data: low on both suggests something else
- ✗ D. 85 percent on training data and 84 percent on test data: a small gap

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-M-Q3 · question page 214

AI.LEARN.04 · Recognize overfitting and good generalization

A model learned that wolf photos have snow and calls a husky in snow a wolf. What happened?

Answer: D. It overfit to an accidental pattern in its photos

Learning an accident of the data (snow) is overfitting.

- ✗ A. Huskies and wolves are the same animal: they are different
- ✗ B. It generalized perfectly from good data: it learned the wrong cue
- ✗ C. It was never trained on any photos of animals at all: it was trained

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-M-Q4 · question page 215

AI.LEARN.04 · Recognize overfitting and good generalization

Which change could reduce overfitting?

Answer: A. Training with more varied examples

Varied data helps a model learn the real pattern.

- ✗ B. Testing on the training data instead: hides the problem
- ✗ C. Removing the validation set: hides the problem
- ✗ D. Training much longer: can make overfitting worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-M-Q5 · question page 216

AI.LEARN.04 · Recognize overfitting and good generalization

Why is overfitting a problem in practice?

Answer: D. The model performs worse on new, real cases

Real use means new examples.

- ✗ A. The training data is deleted automatically: not the problem
- ✗ B. The model refuses to accept any input: not the problem
- ✗ C. The model becomes too fast to measure: not the problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-B1-Q1 · question page 217

AI.LEARN.04 · Compare memorizing answers with understanding the method

Which student overfit?

Answer: A. The one who memorized the practice answers

Memorizing specific answers is overfitting.

- ✗ B. Neither student: the memorizer did
- ✗ C. Both students: only one did
- ✗ D. The one who learned the method for solving them: that student generalized

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-B1-Q2 · question page 218

AI.LEARN.04 · Compare memorizing answers with understanding the method

Why did both students score perfectly on the practice sheet?

Answer: B. Both could answer those exact problems

Familiar examples do not separate memorizing from understanding.

- ✗ A. Both learned the method: only one did
- ✗ C. The practice sheet had no problems: it had 20
- ✗ D. The teacher gave everyone full marks: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-B1-Q3 · question page 219

AI.LEARN.04 · Compare memorizing answers with understanding the method

What happened on the real test with new numbers?

Answer: C. The memorizer struggled; the other did fine

New examples reveal overfitting.

- ✗ A. Both failed: the method-learner succeeded
- ✗ B. The memorizer did better than the method-learner: the reverse
- ✗ D. Both did equally well: they differed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-B1-Q4 · question page 220

AI.LEARN.04 · Compare memorizing answers with understanding the method

Which student is like a model that generalizes?

Answer: B. The method-learner

Generalizing means handling new problems.

- ✗ A. Neither: the method-learner generalizes
- ✗ C. Both: only one generalizes
- ✗ D. The memorizing student: overfit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-B1-Q5 · question page 221

AI.LEARN.04 · Compare memorizing answers with understanding the method

What kind of test reveals overfitting?

Answer: D. A test with new examples

Only unseen examples show the difference.

- ✗ A. A test the student writes: not the point
- ✗ B. A test with no questions: not meaningful
- ✗ C. The same practice sheet again: familiar examples hide it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F1-Q1 · question page 222

AI.LEARN.04 · Diagnose overfitting from results

A model scores 98 percent on training data and 96 percent on validation data. What is the best reading?

Answer: A. It appears to generalize well

A small gap with high scores suggests good generalization.

- ✗ B. It is badly overfit: the gap is small
- ✗ C. It has not learned anything: scores are high
- ✗ D. The validation data must have leaked: no evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F1-Q2 · question page 223

AI.LEARN.04 · Diagnose overfitting from results

A team's model gets 100 percent in training but 60 percent in validation. What should they try?

Answer: C. More varied data or a simpler model

Address the gap with variety or simplicity.

- ✗ A. Stop using any validation data: hides the problem
- ✗ B. Report the 100 percent as its accuracy: misleading
- ✗ D. Train longer on the same data: may worsen overfitting

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F1-Q3 · question page 224

AI.LEARN.04 · Diagnose overfitting from results

A spam filter learned that every email containing "meeting" is safe, because all its safe training emails had that word. What could go wrong?

Answer: D. Spam containing "meeting" will slip through

Overfitting to an accidental cue causes errors on new cases.

- ✗ A. Safe emails will all be blocked: the opposite risk
- ✗ B. Nothing; the word "meeting" proves safety: an accidental pattern
- ✗ C. The filter will stop reading emails: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F1-Q4 · question page 225

AI.LEARN.04 · Diagnose overfitting from results

Which report best shows a model generalizes?

Answer: B. Strong results on new examples it never saw

Generalization is measured on unseen examples.

- ✗ A. A very large number of parameters: size is not evidence
- ✗ C. Perfect results on its own training examples: not enough
- ✗ D. A long training time: time is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F1-Q5 · question page 226

AI.LEARN.04 · Diagnose overfitting from results

A photo model always calls cows "horses" when they stand on sand, because training horses were on beaches. What is the fix?

Answer: A. Train with animals photographed in varied settings

Varied backgrounds break the accidental pattern.

- ✗ B. Increase the model's confidence scores for cows: does not fix learning
- ✗ C. Remove all cow photos from the test: hides the problem
- ✗ D. Show only beach photos from now on: reinforces the bad cue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F2-Q1 · question page 227

AI.LEARN.04 · Tell overfitting from underfitting and a changed world

A model scores 55 percent on training data and 54 percent on validation data for a task where 90 percent is expected. What is the most likely issue?

Answer: C. Underfitting: it has not learned the task well

Low scores on both suggest underfitting.

- ✗ A. It is working perfectly: scores are far below expectations
- ✗ B. The world changed after launch: it has not launched
- ✗ D. Overfitting: training is far higher than validation: they are nearly equal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F2-Q2 · question page 228

AI.LEARN.04 · Tell overfitting from underfitting and a changed world

A fraud model tested well, but months after launch it misses a new kind of scam. What best explains this?

Answer: B. The situation changed in ways its training data did not cover

A changed world can break a model that generalized well before.

- ✗ A. Fraud can never be detected by models: too extreme
- ✗ C. It overfit, since training was far above validation: it tested well on unseen data
- ✗ D. It underfit, since it never learned anything: it tested well

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F2-Q3 · question page 229

AI.LEARN.04 · Tell overfitting from underfitting and a changed world

Which result pattern suggests overfitting?

Answer: A. High training score, much lower validation score

A big gap is the overfitting signature.

- ✗ B. High scores on both, then a drop after launch: a changed world
- ✗ C. Low training score, low validation score: underfitting
- ✗ D. High scores on both with a small gap: good generalization

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F2-Q4 · question page 230

AI.LEARN.04 · Tell overfitting from underfitting and a changed world

A speech model trained on recordings from one microphone does poorly with a new phone model's microphone. What should the team check first?

Answer: D. Whether the new microphone's audio differs from training data

New conditions can differ from training conditions.

✗ A. Whether the old test score was rounded: irrelevant

✗ B. Whether the model has too few parameters to handle speech: not the first check

✗ C. Whether users are speaking too politely: unlikely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S11-F2-Q5 · question page 231

AI.LEARN.04 · Tell overfitting from underfitting and a changed world

A team sees a big training-validation gap and wants to add many more parameters. What is the concern?

Answer: C. A bigger model might memorize more and overfit worse

Added capacity can increase memorization.

✗ A. Validation scores will no longer be measured at all: unrelated

✗ B. More parameters always fix overfitting: not necessarily

✗ D. Parameters cannot be added to models: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-M-Q1 · question page 233

AI.LEARN.05 · Explain how data quality and labels shape a model

A model was trained on product prices from five years ago. Which data problem is this?

Answer: B. Outdated data

Old prices are outdated data.

✗ A. Wrong labels: nothing says labels are wrong

✗ C. Missing data: the data exists but is old

✗ D. Too many parameters: not a data problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-M-Q2 · question page 234

AI.LEARN.05 · Explain how data quality and labels shape a model

Many cat photos in a training set are labeled "dog." What will most likely happen?

Answer: C. The model may confuse cats and dogs

Wrong labels teach wrong answers.

- ✗ A. The model will recognize cats perfectly anyway: wrong labels teach wrong answers
- ✗ B. The model will refuse to train at all: not likely
- ✗ D. Nothing; labels have no effect on learning: labels are what it learns from

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-M-Q3 · question page 235

AI.LEARN.05 · Explain how data quality and labels shape a model

A voice assistant was trained only on adult voices. What problem may appear?

Answer: A. It may understand children poorly

Unrepresentative data leads to weak performance on missing groups.

- ✗ B. It will refuse to hear adult voices: it was trained on them
- ✗ C. It will have no problems at all: unrepresentative data causes gaps
- ✗ D. It will understand children better: unsupported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-M-Q4 · question page 236

AI.LEARN.05 · Explain how data quality and labels shape a model

What is a label in training data?

Answer: D. The correct answer attached to an example

Labels tell the model the answer it should learn.

- ✗ A. A sticker on the user's phone: not a data label
- ✗ B. The title of the finished app: not a data label
- ✗ C. The brand name printed on the computer: not a data label

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-M-Q5 · question page 237

AI.LEARN.05 · Explain how data quality and labels shape a model

What does "garbage in, garbage out" mean for AI?

Answer: B. Problems in the data become problems in the model

A model inherits its data's flaws.

- ✗ A. Models clean up any bad data automatically: they learn whatever the data shows
- ✗ C. Bad data always makes a model refuse to work: it often still works, badly
- ✗ D. Only the model's size affects its quality: data quality matters too

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-B1-Q1 · question page 238

AI.LEARN.05 · Compare learning from a flawed map

A map missing a neighborhood is like which data problem?

Answer: C. Missing data

Absent areas are missing data.

- ✗ A. Unrepresentative data: that is a downtown-only map
- ✗ B. Wrong labels: that is a misnamed street
- ✗ D. Outdated data: that is an old map

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-B1-Q2 · question page 239

AI.LEARN.05 · Compare learning from a flawed map

A street with the wrong name is like which data problem?

Answer: A. Wrong labels

A wrong name is a wrong label.

- ✗ B. Unrepresentative data: not about coverage
- ✗ C. Missing data: the street is there
- ✗ D. Outdated data: the name is simply wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-B1-Q3 · question page 240

AI.LEARN.05 · Compare learning from a flawed map

A twenty-year-old map is like which data problem?

Answer: D. Outdated data

Old information is outdated data.

- ✗ A. Too many streets: not a data problem
- ✗ B. Missing data: not the main issue
- ✗ C. Wrong labels: labels may have been right then

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-B1-Q4 · question page 241

AI.LEARN.05 · Compare learning from a flawed map

A map showing only downtown is like which data problem?

Answer: A. Unrepresentative data

Covering only one part is unrepresentative.

- ✗ B. Wrong labels: the labels may be right
- ✗ C. Outdated data from long ago: it may be current
- ✗ D. Perfect data: it leaves out the suburbs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-B1-Q5 · question page 242

AI.LEARN.05 · Compare learning from a flawed map

Why would you give wrong directions confidently with a flawed map?

Answer: B. You trust what the map shows you

Learned flaws feel like facts.

- ✗ A. You never look at the map: you learned from it
- ✗ C. Directions cannot be wrong: they can
- ✗ D. Maps always tell you: they do not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F1-Q1 · question page 243

AI.LEARN.05 · Identify data problems in real projects

A skin-condition app was trained mostly on photos of light skin. What risk should its makers check?

Answer: C. Weaker accuracy on darker skin

Unrepresentative data can cause uneven accuracy.

- ✗ A. Too many correct answers on light skin: not a risk
- ✗ B. The app running too fast on phones: unrelated
- ✗ D. Users taking photos in good lighting: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F1-Q2 · question page 244

AI.LEARN.05 · Identify data problems in real projects

Volunteers labeled 10,000 tweets as "angry" or "calm" in a rush and made many mistakes. What is the effect?

Answer: D. The model learns some wrong associations

Label errors become model errors.

- ✗ A. The model ignores labels and learns correctly: labels guide learning
- ✗ B. The model will be perfect because 10,000 is many: quantity does not fix wrong labels
- ✗ C. Nothing; rushed labels are always fine: mistakes matter

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F1-Q3 · question page 245

AI.LEARN.05 · Identify data problems in real projects

A rent-prediction model uses listings from 2019 to predict rents today. Which problem is most likely?

Answer: A. Outdated data that misses recent price changes

Old data can miss changes.

- ✗ B. Too much data from today: the opposite
- ✗ C. Wrong labels on every listing: not stated
- ✗ D. Perfect predictions because the data is old: unsupported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F1-Q4 · question page 246

AI.LEARN.05 · Identify data problems in real projects

A traffic model never saw data from snowy days. What could happen on the first snowy day?

Answer: C. Its predictions may be badly wrong

Missing data leaves the model unprepared.

- ✗ A. It will forecast snow by itself: not its job
- ✗ B. Its predictions will be most accurate in snow: unsupported
- ✗ D. Nothing will change at all: missing cases cause errors

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F1-Q5 · question page 247

AI.LEARN.05 · Identify data problems in real projects

Which step best improves the quality of a training set?

Answer: B. Checking a sample of labels for mistakes

Checking labels catches errors.

- ✗ A. Using only the newest computer: not about data quality
- ✗ C. Hiding where all of the data came from: documentation helps quality
- ✗ D. Making the file names shorter: cosmetic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F2-Q1 · question page 248

AI.LEARN.05 · Handle hard data and labeling decisions

Labelers disagree whether "not bad" is positive or neutral. What is the best step?

Answer: D. Write clear labeling instructions and check a sample

Clear instructions and checks make labels consistent.

- ✗ A. Delete every review that is hard to label: loses real cases
- ✗ B. Label every review "positive" to be safe: creates wrong labels

✗ C. Let each labeler decide freely with no guidance: causes inconsistent labels

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F2-Q2 · question page 249

AI.LEARN.05 · Handle hard data and labeling decisions

A team adds 50,000 more photos, but all from the one city already in the data. Why might this not help?

Answer: C. The new data has the same gaps as the old data

More of the same does not fill gaps.

✗ A. Photos from one city cannot be used at all: they can, but coverage stays narrow
✗ B. More data always fixes every problem: not if gaps repeat
✗ D. The model will forget the old photos: not the issue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F2-Q3 · question page 250

AI.LEARN.05 · Handle hard data and labeling decisions

A loan model has no "age" column, but it uses "years since first credit card." Why might age still affect it?

Answer: A. Other data can stand in for the removed column

A related column can act as a stand-in.

✗ B. Credit cards have nothing to do with age: they often relate
✗ C. Models cannot use numbers about time: they can
✗ D. Removing a column always removes its influence: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F2-Q4 · question page 251

AI.LEARN.05 · Handle hard data and labeling decisions

Why should a team document where its data came from and what it leaves out?

Answer: D. So others can judge where the model may fail

Documentation reveals limits.

- ✗ A. So the model trains faster: documentation does not speed training
- ✗ B. So that no one asks any questions about it: the opposite
- ✗ C. So the data file looks longer: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S12-F2-Q5 · question page 252

AI.LEARN.05 · Handle hard data and labeling decisions

A new labeling team is much faster but less careful. What should the project do?

Answer: B. Check a sample of their labels before trusting them

Spot checks reveal label quality.

- ✗ A. Hide their labels from the model: avoids rather than checks
- ✗ C. Trust them, because working fast shows real skill: speed is not accuracy
- ✗ D. Stop labeling data forever: too extreme

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-M-Q1 · question page 254

AI.LEARN.08 · Tell fine-tuning from supplying information in a request

A company trains an existing model further on 20,000 of its own support chats. What is this?

Answer: C. Fine-tuning, which changes the model

Training further on new examples is fine-tuning.

✗ A. Deleting the model's parameters: the parameters are adjusted, not deleted

✗ B. Inference on 20,000 new questions: it is training

✗ D. Supplying information in a request: it is additional training

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-M-Q2 · question page 255

AI.LEARN.08 · Tell fine-tuning from supplying information in a request

Leo pastes his school's rules into his message and asks a question about them. What happens to the model?

Answer: B. Nothing changes; the rules affect only this request

Supplied information affects the request, not the model.

✗ A. It permanently forgets other schools' rules: nothing is deleted

✗ C. It shares the rules with every other user: not how requests work

✗ D. It is fine-tuned on the school rules forever: pasting text does not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-M-Q3 · question page 256

AI.LEARN.08 · Tell fine-tuning from supplying information in a request

Which approach changes the model's parameters?

Answer: D. Fine-tuning

Only further training changes parameters.

✗ A. Typing a longer, clearer question: does not change parameters

✗ B. Attaching a file to one message: does not change parameters

✗ C. Adding three examples to a request: does not change parameters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-M-Q4 · question page 257

AI.LEARN.08 · Tell fine-tuning from supplying information in a request

What does fine-tuning usually require?

Answer: A. Example data, time and some skill

Fine-tuning is a training project.

- ✗ B. Only one sentence typed into a chat: that is a request
- ✗ C. Deleting the original model first: it starts from the existing model
- ✗ D. Nothing; it happens automatically: it is deliberate training

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-M-Q5 · question page 258

AI.LEARN.08 · Tell fine-tuning from supplying information in a request

How long does information supplied in a request affect the model's answers?

Answer: B. Only while it is in that request or conversation

Supplied text acts through the context, not training.

- ✗ A. Exactly one week, then it expires: no such rule
- ✗ C. Until the company retrains the model: it never entered the model
- ✗ D. Forever, for every user of the model everywhere: it does not change the model

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-B1-Q1 · question page 259

AI.LEARN.08 · Compare a training course with a note on the desk

Fine-tuning is like which option?

Answer: C. The weeks-long training course

Further training changes the learner.

- ✗ A. Both options: only the course changes her
- ✗ B. The note left on the desk: that is supplying information
- ✗ D. Neither option: it matches the course

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-B1-Q2 · question page 260

AI.LEARN.08 · Compare a training course with a note on the desk

Pasting information into a request is like which option?

Answer: A. The note on the desk

A note helps the current task.

- ✗ B. Hiring a new employee: not the comparison
- ✗ C. The training course: that is fine-tuning
- ✗ D. Closing the store: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-B1-Q3 · question page 261

AI.LEARN.08 · Compare a training course with a note on the desk

Does the note change what the employee learned in training?

Answer: D. No, it only helps today

Notes do not change training.

- ✗ A. No, because she cannot read: not the point
- ✗ B. Yes, it replaces the course: it does not
- ✗ C. Yes: a note is temporary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-B1-Q4 · question page 262

AI.LEARN.08 · Compare a training course with a note on the desk

Which lasts after the task is done?

Answer: A. What she learned in the course

Training lasts; notes are temporary.

- ✗ B. Neither: the course lasts
- ✗ C. The note's message: it was for today
- ✗ D. Only the store hours in the note: those change

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-B1-Q5 · question page 263

AI.LEARN.08 · Compare a training course with a note on the desk

Tomorrow the store closes at 8. What is the easiest way to tell her?

Answer: C. Leave a new note

Changing facts are easiest to supply as notes.

- ✗ A. Tell no one: she needs to know
- ✗ B. Hire a different employee: unnecessary
- ✗ D. Send her back to a weeks-long course: too slow for a changing fact

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F1-Q1 · question page 264

AI.LEARN.08 · Choose between fine-tuning and supplying information

A small shop wants answers about its return policy, which it updates often. Which approach is simplest to keep current?

Answer: B. Supply the current policy in each request

Supplying current text is easy to update.

- ✗ A. Hope the model already knows the shop's policy: it likely does not
- ✗ C. Fine-tune a model every time the policy changes: slow and costly
- ✗ D. Fine-tune once and never update it: it will go out of date

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F1-Q2 · question page 265

AI.LEARN.08 · Choose between fine-tuning and supplying information

Tia attached a recipe file and asked for a shopping list. Did the model learn her recipe permanently?

Answer: D. No; the file was used only for that request

An attachment adds context, not training.

- ✗ A. Yes; attaching a file fine-tunes the model: attaching does not retrain
- ✗ B. Yes; all users can now see her recipe: not how it works
- ✗ C. No; models cannot read attached files: many products can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F1-Q3 · question page 266

AI.LEARN.08 · Choose between fine-tuning and supplying information

A law firm wants a model to always write in its strict house format across thousands of documents. Which approach might suit it?

Answer: C. Fine-tuning on many examples of that format

A lasting style across many uses can suit fine-tuning, with testing.

- ✗ A. Never giving any examples: examples are needed
- ✗ B. Typing "use our format" once and assuming it lasts forever: requests do not persist in the model
- ✗ D. Asking a different model each time: inconsistent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F1-Q4 · question page 267

AI.LEARN.08 · Choose between fine-tuning and supplying information

Which statement about fine-tuning is accurate?

Answer: A. It changes the model for later uses of that version

Fine-tuning creates a changed version.

- ✗ B. It needs no data at all: it needs examples
- ✗ C. It happens whenever a user says "learn this": it is deliberate training
- ✗ D. It changes only the current message: that is supplying information

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F1-Q5 · question page 268

AI.LEARN.08 · Choose between fine-tuning and supplying information

A teacher wants a model to answer questions about this week's homework sheet. What is the easiest approach?

Answer: D. Include the homework sheet in the request

Supply the sheet directly.

- ✗ A. Ask the model to guess the sheet: unreliable
- ✗ B. Expect the model to know the sheet already: it has never seen it
- ✗ C. Fine-tune a model on the sheet: too much effort for one week

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F2-Q1 · question page 269

AI.LEARN.08 · Decide when fine-tuning is the wrong tool

A store wants a model to know its 2,000 prices, which change weekly. Why is fine-tuning a poor

fit?

Answer: B. Prices change weekly; fine-tuning suits that poorly

Changing facts are better supplied from a current source.

- ✗ A. Prices are illegal to share with any AI model: not generally
- ✗ C. Fine-tuning works only for images: not true
- ✗ D. Fine-tuning cannot use numbers at all: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F2-Q2 · question page 270

AI.LEARN.08 · Decide when fine-tuning is the wrong tool

After fine-tuning, a model starts making errors it did not make before. What does this show?

Answer: A. Fine-tuning can add new mistakes, so test the result

Any changed version needs testing.

- ✗ B. The original model was deleted by accident: not implied
- ✗ C. Fine-tuning always makes every answer better: not guaranteed
- ✗ D. Errors cannot happen after fine-tuning: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F2-Q3 · question page 271

AI.LEARN.08 · Decide when fine-tuning is the wrong tool

A team wants to fine-tune on customer emails that include home addresses. What should they consider first?

Answer: B. Whether they may use that private data, and how to protect it

Fine-tuning data needs the same care as any data sharing.

- ✗ A. Nothing; private data is fine to use for any kind of training: requires care

and permission

✗ C. Only how fast the training will finish: privacy comes first

✗ D. Whether the emails are long enough: not the main concern

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F2-Q4 · question page 272

AI.LEARN.08 · Decide when fine-tuning is the wrong tool

Which task is the best fit for supplying information in the request rather than fine-tuning?

Answer: C. Answering questions about this month's updated policy

Current, changing information is easiest to supply.

✗ A. Creating a new version for long-term use: fine-tuning

✗ B. Training a model on a large specialized task: fine-tuning may suit

✗ D. Teaching a lasting writing style used across all documents: could suit fine-tuning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S13-F2-Q5 · question page 273

AI.LEARN.08 · Decide when fine-tuning is the wrong tool

A manager says, "We fine-tuned it, so we no longer need to check its answers." What is wrong?

Answer: D. Tuned models still make mistakes and need testing

Testing is needed either way.

✗ A. Fine-tuning deletes the ability to check: nonsense

✗ B. Checking is only for models that were never tuned: all models need checking

✗ C. Nothing; fine-tuned models are always correct: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-M-Q1 · question page 275

AI.LLM.01 · Describe what a large language model does

What is the core job a large language model is trained to do?

Answer: B. Predict a likely next piece of text

An LLM predicts likely next text, repeatedly.

✗ A. Store a copy of the internet to read from: it is a trained model, not a stored copy

✗ C. Look up each answer in a database of verified facts: it predicts text; it does not check facts

✗ D. Follow a fixed rule list written by programmers: it learned patterns from text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-M-Q2 · question page 276

AI.LLM.01 · Describe what a large language model does

What does "large" refer to in "large language model"?

Answer: C. The size of the model and its training data

"Large" describes model size and training data.

✗ A. The number of languages printed on its box: not what "large" means

✗ B. The length of the answers it is allowed to write: not what "large" means

✗ D. The physical size of the computer screen it uses: not what "large" means

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-M-Q3 · question page 277

AI.LLM.01 · Describe what a large language model does

How does an LLM produce a whole paragraph?

Answer: A. By repeating next-piece predictions one after another

Repeated prediction builds longer text.

- ✗ B. By copying one paragraph from a single website: it generates piece by piece
- ✗ C. By asking a human writer to type each sentence: it generates automatically
- ✗ D. By choosing a finished paragraph from a stored list: it builds text step by step

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-M-Q4 · question page 278

AI.LLM.01 · Describe what a large language model does

Why can an LLM's answer be plausible but wrong?

Answer: D. Likely-sounding text is not always true

Prediction favors plausible text, which may be false.

- ✗ A. It is programmed to make mistakes on purpose: errors are not deliberate
- ✗ B. It writes only random words with no pattern: its output follows patterns
- ✗ C. It always checks facts before it writes: it has no built-in truth check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-M-Q5 · question page 279

AI.LLM.01 · Describe what a large language model does

Which task is an LLM usually well suited to help with?

Answer: C. Drafting a first version of a short letter

Drafting text plays to an LLM's strengths; facts still need checks.

✗ A. Certifying that a medicine dose is safe for you: needs a qualified professional

✗ B. Proving a fact is true just by stating it: stating is not proving

✗ D. Guaranteeing the exact current price of a stock: needs a live, checked source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-B1-Q1 · question page 280

AI.LLM.01 · Compare an LLM with a phone's word suggestions

After "See you," why might a phone suggest "soon"?

Answer: B. "Soon" often comes next in messages

Suggestions reflect common patterns.

✗ A. It knows when you will meet: it does not know your plans

✗ C. It was told by your friend: not how suggestions work

✗ D. It checked your calendar for the date: suggestions come from patterns

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-B1-Q2 · question page 281

AI.LLM.01 · Compare an LLM with a phone's word suggestions

What core job do the phone keyboard and an LLM share?

Answer: D. Predicting likely next text

Both predict likely continuations.

✗ A. Taking photos of the screen: unrelated

✗ B. Sending messages for you: unrelated

✗ C. Checking facts against a database: neither does this by default

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-B1-Q3 · question page 282

AI.LLM.01 · Compare an LLM with a phone's word suggestions

What makes an LLM much more capable than a phone keyboard?

Answer: A. Its far bigger scale of training

Scale greatly increases ability.

- ✗ B. It only works offline: not the reason
- ✗ C. It uses fewer words: not the reason
- ✗ D. It has a much larger screen: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-B1-Q4 · question page 283

AI.LLM.01 · Compare an LLM with a phone's word suggestions

Does a likely next word have to be true for you?

Answer: C. No, likely is not the same as true

Likely text can be wrong for your situation.

- ✗ A. No, because phones cannot spell: not the point
- ✗ B. Yes, likely words are always true: not necessarily
- ✗ D. Yes, if the phone is expensive: price does not matter

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-B1-Q5 · question page 284

AI.LLM.01 · Compare an LLM with a phone's word suggestions

Which continuation would a keyboard most likely suggest after "Happy"?

Answer: A. "birthday"

Common phrases are likely continuations.

✗ B. "tax refund deadline": unlikely continuation

✗ C. "spreadsheet": rarely follows "Happy"

✗ D. "carburetor": unlikely continuation

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F1-Q1 · question page 285

AI.LLM.01 · Apply what an LLM is to everyday uses

Rosa asks an LLM-based chatbot for the population of her town and gets a precise number. What should she do before using it in a report?

Answer: B. Check the number in an official source

A plausible number still needs a source.

✗ A. Round it up so it looks more official: does not check it

✗ C. Ask the chatbot to repeat the same number: repetition is not a check

✗ D. Use it: precision is not accuracy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F1-Q2 · question page 286

AI.LLM.01 · Apply what an LLM is to everyday uses

A student wants help turning bullet notes into a clear paragraph. Is an LLM a reasonable tool to try?

Answer: D. Yes; rewriting suits it, then check the facts

Drafting suits LLMs; review keeps it accurate.

✗ A. No; LLMs cannot produce paragraphs: they can

✗ B. Yes; and the result needs no checking: facts and meaning still need review

✗ C. No; only search engines can rewrite notes: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S14-F1-Q3 · question page 287

AI.LLM.01 · Apply what an LLM is to everyday uses

Which description of an LLM is most accurate for a beginner?

Answer: B. A model that writes by predicting text, which can be wrong

Prediction-based writing with possible errors.

- ✗ A. A search engine that only shows links: it generates text
- ✗ C. A robot that understands everything just like a person: misleading comparison
- ✗ D. A talking encyclopedia that is always correct: it can be wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F1-Q4 · question page 288

AI.LLM.01 · Apply what an LLM is to everyday uses

An LLM writes a confident paragraph about a law. What is the right attitude?

Answer: C. Treat it as a draft and check the law's actual text

Useful draft, unverified facts.

- ✗ A. Ignore it completely, since LLMs never help: it may help as a starting point
- ✗ B. Trust it, because legal wording sounds official: tone is not evidence
- ✗ D. Assume it is current because it sounds confident: it may be outdated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F1-Q5 · question page 289

AI.LLM.01 · Apply what an LLM is to everyday uses

Why might two well-written LLM answers about the same history event disagree?

Answer: A. Each is predicted text; either could contain errors

Plausible text can differ and be wrong; check a source.

- ✗ B. History has no facts that can be checked: it does
- ✗ C. LLMs always agree on facts: they do not
- ✗ D. One of them must have been copied from a textbook: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F2-Q1 · question page 290

AI.LLM.01 · Avoid both extreme claims about LLMs

A friend says, "LLMs are just autocomplete, so they are useless." What is the best response?

Answer: D. Prediction at large scale can still be very useful

The "useless" slogan ignores real abilities.

- ✗ A. Correct; they cannot help with any real task: they help with drafting and more
- ✗ B. Correct; autocomplete never produces sentences: it can
- ✗ C. Wrong; LLMs never predict text at all: prediction is their core job

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F2-Q2 · question page 291

AI.LLM.01 · Avoid both extreme claims about LLMs

A coworker says, "It knows everything, so I never check it." What is the risk?

Answer: A. Confident but wrong answers will slip through

The "knows everything" slogan hides errors.

- ✗ B. The model will stop working if not checked: not how it works
- ✗ C. Checking is illegal for business tools: nonsense
- ✗ D. None; LLMs cannot make errors: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F2-Q3 · question page 292

AI.LLM.01 · Avoid both extreme claims about LLMs

What is the best way to judge whether an LLM is good for your task?

Answer: B. Test it on examples of your task and check the facts

Testing beats slogans.

- ✗ A. Count how many people use it worldwide: popularity is not fit
- ✗ C. Ask it whether it is good at your task: self-reports are not tests
- ✗ D. Decide based on a slogan about what it "really" is: slogans are not tests

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F2-Q4 · question page 293

AI.LLM.01 · Avoid both extreme claims about LLMs

An LLM summarized a report well but added a statistic not in the report. What does this show?

Answer: D. It can be useful and still make confident errors

Usefulness and errors can coexist.

- ✗ A. It is always right about statistics: it added one
- ✗ B. Summaries never contain errors: this one did
- ✗ C. It is useless for every kind of task at all: it summarized well

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S14-F2-Q5 · question page 294

AI.LLM.01 · Avoid both extreme claims about LLMs

Why can a likely-sounding sentence be false?

Answer: C. Prediction favors typical-sounding text, not checked text

Typical-sounding is not the same as verified.

- ✗ A. Likely sentences are always checked against sources first: no built-in check
- ✗ B. LLMs cannot write sentences: they can
- ✗ D. False sentences are always unlikely: false claims can sound typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S15-M-Q1 · question page 296

AI.LLM.02 · Explain tokens and why counts differ from words

What is a token?

Answer: A. A chunk of text, such as a word or part of one

Tokens are pieces of text.

- ✗ B. A coin you pay with to use an AI website: not the meaning here
- ✗ C. A password that unlocks a model: not the meaning here
- ✗ D. A whole page of text read at once: tokens are small chunks

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-M-Q2 · question page 297

AI.LLM.02 · Explain tokens and why counts differ from words

Which word is (a very common short word) most likely split into several tokens?

Answer: C. unbelievably (a long, rarer word)

Long or rare words are often split.

- ✗ A. the (a very common short word): a common short word
- ✗ B. and (a very common short word): a common short word
- ✗ D. is (a very common short word): a common short word

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-M-Q3 · question page 298

AI.LLM.02 · Explain tokens and why counts differ from words

Using the rule of thumb of about three quarters of a word per token, about how many tokens is a 300-word English text?

Answer: B. About 400 tokens

300 / 0.75 = 400 tokens (an estimate).

- ✗ A. About 225 tokens: that multiplies by 0.75 instead of dividing
- ✗ C. About 1,200 tokens: that is the character estimate
- ✗ D. About 300 tokens: tokens are not equal to words

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-M-Q4 · question page 299

AI.LLM.02 · Explain tokens and why counts differ from words

Why do token counts matter to users?

Answer: D. Many limits and prices are measured in tokens

Limits and usage prices often use tokens.

- ✗ A. Tokens decide whether the answer is true: unrelated
- ✗ B. Tokens are only used for counting images: they measure text
- ✗ C. Tokens decide the font the answer appears in: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-M-Q5 · question page 300

AI.LLM.02 · Explain tokens and why counts differ from words

Is the token rule of thumb exact for every model?

Answer: A. No; counts depend on how each model splits text

Rules of thumb are estimates.

- ✗ B. No; tokens are always exactly one letter: not true
- ✗ C. Yes; one token always equals one word: not true
- ✗ D. Yes; every model splits text identically: models differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-B1-Q1 · question page 301

AI.LLM.02 · Estimate tokens from words

About how many tokens is 30 words?

Answer: B. 40

30 / 0.75 = 40.

- ✗ A. 22.5: multiplied instead of dividing
- ✗ C. 120: used the character estimate
- ✗ D. 30: tokens are not words

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-B1-Q2 · question page 302

AI.LLM.02 · Estimate tokens from words

About how many words is 400 tokens?

Answer: C. 300

400 x 0.75 = 300.

- ✗ A. 400: words are not tokens
- ✗ B. 533: divided instead of multiplying
- ✗ D. 100: used the wrong factor

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-B1-Q3 · question page 303

AI.LLM.02 · Estimate tokens from words

About how many tokens is 150 words?

Answer: D. 200

150 / 0.75 = 200.

- ✗ A. 112.5: multiplied instead of dividing
- ✗ B. 150: tokens are not words
- ✗ C. 600: used the character estimate

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-B1-Q4 · question page 304

AI.LLM.02 · Estimate tokens from words

Dividing by 0.75 gives the same result as which steps?

Answer: A. Multiply by 4, then divide by 3

x / 0.75 = x x 4 / 3.

- ✗ B. Add 75 to the number: not equivalent
- ✗ C. Subtract a quarter: not equivalent
- ✗ D. Multiply by 3, then divide by 4: that is multiplying by 0.75

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-B1-Q5 · question page 305

AI.LLM.02 · Estimate tokens from words

About how many words is 1,000 tokens?

Answer: D. 750

1,000 x 0.75 = 750.

- ✗ A. 250: subtracted instead
- ✗ B. 1,333: divided instead of multiplying
- ✗ C. 1,000: words are not tokens

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F1-Q1 · question page 306

AI.LLM.02 · Use token estimates in practical situations

A tool allows about 2,000 tokens per message. About how many words of English is that?

Answer: C. About 1,500 words

2,000 x 0.75 = 1,500 words (an estimate).

- ✗ A. About 2,667 words: divided instead of multiplying
- ✗ B. About 500 words: subtracted a quarter of a quarter
- ✗ D. Exactly 2,000 words: tokens are not words

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F1-Q2 · question page 307

AI.LLM.02 · Use token estimates in practical situations

Jun's essay is 900 words. About how many tokens will it use?

Answer: B. About 1,200 tokens

900 / 0.75 = 1,200 tokens (an estimate).

- ✗ A. Exactly 900 tokens: tokens are not words
- ✗ C. About 3,600 tokens: used the character estimate
- ✗ D. About 675 tokens: multiplied instead of dividing

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F1-Q3 · question page 308

AI.LLM.02 · Use token estimates in practical situations

Why might a message hit a token limit even though it seems short in words?

Answer: A. Some words and symbols use several tokens each

Rare words, numbers and symbols can use more tokens.

- ✗ B. Token limits only count pictures: they count text
- ✗ C. Every word always uses exactly ten tokens: not true
- ✗ D. Spaces are never counted at all: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F1-Q4 · question page 309

AI.LLM.02 · Use token estimates in practical situations

A pricing page charges by tokens. What does that mean for a long document?

Answer: C. Longer text usually costs more tokens

More text means more tokens, usually more cost.

- ✗ A. Long documents are always free: not implied
- ✗ B. Cost depends on the font size used: unrelated
- ✗ D. Cost depends only on the number of pages printed: tokens, not pages

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F1-Q5 · question page 310

AI.LLM.02 · Use token estimates in practical situations

Two tools report different token counts for the same paragraph. What is the likely reason?

Answer: D. They split text into tokens differently

Different tokenizers give different counts.

- ✗ A. One tool must be broken: differences are normal
- ✗ B. Tokens are random numbers: they follow rules
- ✗ C. Paragraphs have no fixed text inside them: the text is the same

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F2-Q1 · question page 311

AI.LLM.02 · Handle surprising token counts

A message with a long table of numbers uses many more tokens than a story of the same word count. Why?

Answer: B. Numbers and table symbols often take more tokens

Long numbers and symbols raise token counts.

- ✗ A. Tables are always converted into images first: not implied
- ✗ C. Numbers always count as zero tokens: the opposite
- ✗ D. Stories are never counted in tokens: they are

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F2-Q2 · question page 312

AI.LLM.02 · Handle surprising token counts

A fluent model miscounts how many times the letter r appears in "strawberry." What helps explain this?

Answer: C. It works with tokens, not individual letters

Seeing chunks, not letters, can trip up letter tasks.

- ✗ A. It cannot read English at all: it writes fluent English
- ✗ B. Letter counting is always easy for every model: not when it sees tokens

✗ D. The word is spelled differently for computers: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F2-Q3 · question page 313

AI.LLM.02 · Handle surprising token counts

You estimate a cost using one tool's token counter, but you will use a different model. What should you remember?

Answer: A. The count is only an estimate for the other model

Different models tokenize differently.

✗ B. Token counts never affect cost: they often do

✗ C. You must count tokens by hand instead: a counter is still a useful estimate

✗ D. The count will match exactly for every model: models split differently

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S15-F2-Q4 · question page 314

AI.LLM.02 · Handle surprising token counts

A message in a language other than English uses more tokens than its English translation. What is the best explanation?

Answer: D. Some languages use more tokens per word for some models

Token efficiency varies by language and model.

✗ A. Tokens count only English letters: not true

✗ B. Languages other than English are not allowed to be tokenized: not true

✗ C. The translation must have deleted words: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

S15-F2-Q5 · question page 315

AI.LLM.02 · Handle surprising token counts

A code snippet of 50 lines seems short but uses a lot of tokens. What is the likely reason?

Answer: B. Symbols, spaces and names in code split into many tokens

Code can be token-heavy.

- ✗ A. Each line of code always counts as exactly one token: not true
- ✗ C. Code is turned into pictures first: not implied
- ✗ D. Code is never counted in tokens: it is

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: tokens_openai

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S16-M-Q1 · question page 317

AI.LLM.03 · Explain probabilities and why answers vary

What does a language model assign to each possible next token?

Answer: A. A probability of how likely it is to come next

Each candidate token gets a probability.

- ✗ B. A price for using that particular token: not what it assigns
- ✗ C. A fixed rule saying which token is required: it weighs options
- ✗ D. A grade showing whether it is true or false: probability is not truth

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-M-Q2 · question page 318

AI.LLM.03 · Explain probabilities and why answers vary

Using the table, which token is the most likely next choice?

Prepared example (made-up probabilities for teaching)

Possible next token after "The sky is"	Probability (made-up numbers)

blue	0.62
clear	0.18
gray	0.12
green	0.01
all other tokens	0.07

Answer: D. blue

blue has the highest probability (0.62).

- ✗ A. clear: 0.18 is lower than 0.62
- ✗ B. gray: 0.12 is lower
- ✗ C. green: 0.01 is very low

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-M-Q3 · question page 319

AI.LLM.03 · Explain probabilities and why answers vary

Using the table, what is the combined probability of "clear" or "gray"?

Prepared example (made-up probabilities for teaching)

Possible next token after "The sky is"	Probability (made-up numbers)
blue	0.62
clear	0.18
gray	0.12
green	0.01
all other tokens	0.07

Answer: B. 0.30

0.18 + 0.12 = 0.30.

- ✗ A. 0.80: added "blue" and "clear" instead
- ✗ C. 0.18: only "clear"

✗ D. 0.12: only "gray"

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S16-M-Q4 · question page 320

AI.LLM.03 · Explain probabilities and why answers vary

Why can the same request produce different answers on different tries?

Answer: C. The model often samples among likely tokens

Weighted random choices lead to variation.

✗ A. Someone edits the answers by hand each time: not how it works
✗ B. The internet changes the model every second: not the usual reason
✗ D. The model forgets its training each time: its training stays the same

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-M-Q5 · question page 321

AI.LLM.03 · Explain probabilities and why answers vary

Settings such as "temperature" usually change what?

Answer: D. How predictable or varied the choices are

These settings adjust variation.

✗ A. Whether the facts are true: settings do not check truth
✗ B. How warm the computer runs: not this meaning
✗ C. Which language the user speaks and reads: not this setting

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-B1-Q1 · question page 322

AI.LLM.03 · Read and combine simple probabilities

Which probability means "certain"?

Answer: B. 1

1 means certain.

- ✗ A. 0.5: as likely as not
- ✗ C. 0.1: unlikely
- ✗ D. 0: impossible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-B1-Q2 · question page 323

AI.LLM.03 · Read and combine simple probabilities

Rain 0.3, snow 0.1. What is the chance of rain or snow?

Answer: C. 0.4

0.3 + 0.1 = 0.4.

- ✗ A. 0.2: subtracted instead
- ✗ B. 0.03: multiplied instead
- ✗ D. 0.3: only rain

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S16-B1-Q3 · question page 324

AI.LLM.03 · Read and combine simple probabilities

Three options have probabilities 0.5, 0.3 and a missing value. What is the missing value?

Answer: A. 0.2

1 - 0.5 - 0.3 = 0.2.

- ✗ B. 1.0: that is the total
- ✗ C. 0.8: added instead of finding what is left

✗ D. 0.5: repeated a value

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S16-B1-Q4 · question page 325

AI.LLM.03 · Read and combine simple probabilities

Which option is most likely: A 0.25, B 0.45, C 0.30?

Answer: B. B

B has the highest probability.

✗ A. A: 0.25 is lowest
✗ C. All are equal: they differ
✗ D. C: 0.30 is less than 0.45

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-B1-Q5 · question page 326

AI.LLM.03 · Read and combine simple probabilities

An option has probability 0.9. Is it guaranteed to happen?

Answer: A. No, it is likely but not certain

Likely is not guaranteed.

✗ B. Yes, 0.9 equals 1: it does not
✗ C. No, it is impossible: 0 is impossible
✗ D. Yes, anything over 0.5 is certain: only 1 is certain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F1-Q1 · question page 327

AI.LLM.03 · Apply variation to everyday use

Ali asks for a party theme twice and gets two different ideas. What does this most likely mean?

Answer: D. The model sampled differently; both are normal

For ideas, variation is expected and useful.

- ✗ A. Only one of the ideas could be a real party theme: both can be valid ideas
- ✗ B. His first request was deleted by mistake: not implied
- ✗ C. The model is broken and should be reported: variation is normal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F1-Q2 · question page 328

AI.LLM.03 · Apply variation to everyday use

A teacher wants many different practice sentences. What does variation offer?

Answer: C. Several different options from similar requests

Variation is a feature for brainstorming.

- ✗ A. A guarantee that every sentence is correct: variation does not guarantee correctness
- ✗ B. The same sentence every single time: the opposite
- ✗ D. Nothing useful for this task: variety helps here

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F1-Q3 · question page 329

AI.LLM.03 · Apply variation to everyday use

A tool offers a setting for "more creative" or "more focused" answers. Which choice fits writing a list of unusual story ideas?

Answer: A. More creative (more varied)

Varied output helps brainstorming.

✗ B. More focused (more predictable): less variety for ideas

✗ C. It makes no difference at all: it changes variation

✗ D. Turn the tool off: avoids the task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F1-Q4 · question page 330

AI.LLM.03 · Apply variation to everyday use

Why is a likely next token not guaranteed to be correct?

Answer: B. Likely means common in patterns, not checked as true

Probability reflects patterns, not truth.

✗ A. Likely tokens are always true facts: not necessarily

✗ C. Correct tokens always get a probability of exactly 1: not how it works

✗ D. Probabilities are only used for pictures: they are used for text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F1-Q5 · question page 331

AI.LLM.03 · Apply variation to everyday use

A user wants three different subject lines for an email. What is the best request?

Answer: C. Ask for three distinct options in one request

Asking for several options uses variation on purpose.

✗ A. Ask for exactly one line and copy it three times: not distinct

✗ B. Never use AI for subject lines: unnecessary

✗ D. Ask once and expect the same line every time: one line is not three options

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F2-Q1 · question page 332

AI.LLM.03 · Respond sensibly when answers vary

You ask a model the height of a famous tower twice and get 324 meters and 330 meters. What should you do?

Answer: D. Check a reliable source; at most one is right

Different answers to a factual question signal the need to check.

- ✗ A. Ask a third time and trust whichever number repeats: agreement is not proof
- ✗ B. Average them to 327 and report that: averaging guesses is not a check
- ✗ C. Pick the number you like better: preference is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F2-Q2 · question page 333

AI.LLM.03 · Respond sensibly when answers vary

A store needs 100 product descriptions in exactly the same format. What helps most?

Answer: B. Clear instructions, a template, and checking the results

Templates and checks support consistency.

- ✗ A. Assuming outputs are always identical: they may differ
- ✗ C. Higher variation settings for more creativity: works against consistency
- ✗ D. Asking without any format and hoping it matches: unlikely to be consistent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F2-Q3 · question page 334

AI.LLM.03 · Respond sensibly when answers vary

Even with "predictable" settings, a product gives slightly different wording each time. What is the

best conclusion?

Answer: D. Settings reduce variation but may not guarantee identical output

Do not count on identical outputs.

- ✗ A. The answers are therefore all false: wording differences do not mean false
- ✗ B. The settings must be ignored completely by every product: too strong
- ✗ C. The model is learning from each request: ordinary use does not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F2-Q4 · question page 335

AI.LLM.03 · Respond sensibly when answers vary

When is variation in answers most helpful?

Answer: A. When you want many ideas to choose from

Variety helps idea generation.

- ✗ B. When you need a precise measurement: needs a source
- ✗ C. When you need one exact legal date: needs a source, not variety
- ✗ D. When you must match a strict template: variety works against it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S16-F2-Q5 · question page 336

AI.LLM.03 · Respond sensibly when answers vary

Two runs of the same request give different conclusions about whether a contract clause is risky. What should a careful user do?

Answer: C. Treat it as uncertain and get the clause reviewed

Conflicting answers signal uncertainty; get a reliable check.

✗ A. Run it ten times and take the most common answer: agreement is not proof

✗ B. Ignore the clause entirely: unsafe

✗ D. Choose the more reassuring conclusion: preference is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-M-Q1 · question page 338

AI.LLM.04 · Explain why fluent language is not evidence

Why do language model answers usually sound smooth and confident?

Answer: D. They are trained to produce natural-sounding text

Fluency comes from training on natural text.

✗ A. They have checked every fact before writing: fluency does not come from checking

✗ B. They copy only from verified textbooks: not how they work

✗ C. They are always written by human experts: they are generated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-M-Q2 · question page 339

AI.LLM.04 · Explain why fluent language is not evidence

In this fictional example, the book title was invented for the question. What does the response show?

Fictional example (prepared in advance; not a real AI output)

Request:

Who wrote the novel "The Silver Lantern of Brightwater"?

Response:

The novel "The Silver Lantern of Brightwater" was written by Eleanor

Marsh and published in 1962. It won praise for its vivid portrait of a fishing town and is often taught in literature classes.

Answer: A. A confident, detailed answer about something unreal

Fluent detail can describe something unreal.

- ✗ B. That the model checked a library catalog: nothing shows a check
- ✗ C. That any answer with a year must be correct: a year is not evidence
- ✗ D. Proof that the book exists, since details are given: details are not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-M-Q3 · question page 340

AI.LLM.04 · Explain why fluent language is not evidence

An answer says, "Studies show 73 percent of people prefer this." What does that sentence prove by itself?

Answer: C. Nothing yet; you would need the actual studies

"Studies show" is a claim, not evidence.

- ✗ A. That a real study found exactly 73 percent: the phrase is not a source
- ✗ B. That the number is rounded correctly: not established
- ✗ D. That most people prefer it, for certain: not established

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-M-Q4 · question page 341

AI.LLM.04 · Explain why fluent language is not evidence

Which of these is something you can actually check?

Answer: B. A link to the report that you can open

A source you can open can be checked.

✗ A. A very precise-looking number with decimals: precision is not proof

✗ C. A confident, formal tone: tone is not checkable evidence

✗ D. The phrase "experts agree": vague, not a source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-M-Q5 · question page 342

AI.LLM.04 · Explain why fluent language is not evidence

A model gives a correct-sounding rule for a game. What is the best way to confirm it?

Answer: D. Check the official rulebook

An official source confirms a rule.

✗ A. Ask the model whether it is sure: its answer is more generated text

✗ B. Trust it because it is well organized: organization is not evidence

✗ C. Assume it is right since it is fluent: fluency is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-B1-Q1 · question page 343

AI.LLM.04 · Separate how a claim sounds from what supports it

Which part tells you whether to trust a claim?

Answer: B. What supports it

Support decides trust.

✗ A. How long it is: length is not support

✗ C. How formal the words are: style is not support

✗ D. How confident it sounds: sound is not support

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-B1-Q2 · question page 344

AI.LLM.04 · Separate how a claim sounds from what supports it

Which claim includes support you can check?

Answer: C. "Opens at 9 a.m., per its website (link)"

A link can be checked.

- ✗ A. "Everyone knows it opens at 9": vague
- ✗ B. "Definitely opens at 9 a.m.": only sounds sure
- ✗ D. "It surely opens at 9 a.m., just trust me": only sounds sure

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-B1-Q3 · question page 345

AI.LLM.04 · Separate how a claim sounds from what supports it

"The bridge is 2.3 km long, I am completely certain." What kind of statement is "I am completely certain"?

Answer: A. How it sounds, not support

Certainty words are about sound.

- ✗ B. A source that you can open: nothing to open
- ✗ C. An official record: not a record
- ✗ D. A calculation: no calculation

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-B1-Q4 · question page 346

AI.LLM.04 · Separate how a claim sounds from what supports it

Can a claim sound unsure and still be true?

Answer: C. Yes; how it sounds does not decide truth

Sound and truth are separate.

✗ A. No; true claims always sound sure: not true

✗ B. Only if it is very long: length is irrelevant

✗ D. No; unsure claims are always false: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-B1-Q5 · question page 347

AI.LLM.04 · Separate how a claim sounds from what supports it

Which is a calculation you can redo?

Answer: D. "3 tickets at 12 dollars each is 36 dollars"

You can redo 3 x 12.

✗ A. "The price is surely fair": opinion

✗ B. "Everyone online says the tickets are cheap": hearsay

✗ C. "The tickets are a great deal": opinion

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F1-Q1 · question page 348

AI.LLM.04 · Apply "fluent is not evidence" to real answers

A chatbot gives a detailed biography of a local artist, including awards. Nothing else online mentions them. What is the most reasonable conclusion?

Answer: A. Some details may be invented; confirm them elsewhere

Unconfirmed detail may be invented.

✗ B. Search engines must be hiding the awards: unsupported guess

✗ C. The biography must be true because it is detailed: detail is not proof

✗ D. The artist must be famous but very private: unsupported guess

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F1-Q2 · question page 349

AI.LLM.04 · Apply "fluent is not evidence" to real answers

Sara needs a statistic for a speech. The model gives one with no source. What should she do?

Answer: B. Find the original source before using it

A statistic needs a checkable source.

- ✗ A. Use it but round it to sound safer: still unchecked
- ✗ C. Ask the model to sound more confident: confidence is not evidence
- ✗ D. Use it, because it looks precise: precision is not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F1-Q3 · question page 350

AI.LLM.04 · Apply "fluent is not evidence" to real answers

An answer explains a tax rule in formal language. Which step best checks it?

Answer: C. Compare it with the official government page

An official source is the check.

- ✗ A. Read it again to see if it still sounds right: sounding right is not checking
- ✗ B. Trust it because tax rules are simple: they often are not
- ✗ D. Check whether it uses legal vocabulary: vocabulary is not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F1-Q4 · question page 351

AI.LLM.04 · Apply "fluent is not evidence" to real answers

Which answer feature is the best sign you can verify a claim?

Answer: D. It names a specific source you can look up

A specific, checkable source supports verification.

✗ A. It is longer than other answers: not evidence

✗ B. It uses many technical words: not evidence

✗ C. It is written in a calm, confident tone: not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F1-Q5 · question page 352

AI.LLM.04 · Apply "fluent is not evidence" to real answers

A model confidently states a store's holiday hours. What is the most reliable next step?

Answer: B. Check the store's own website or call it

Go to the source that controls the fact.

✗ A. Assume all stores have the same hours: unsupported

✗ C. Ask the model to repeat it: repetition is not a check

✗ D. Trust it because it sounded sure: not evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F2-Q1 · question page 353

AI.LLM.04 · Check the claims that matter in a mixed answer

An answer ends with "This is a verified fact." What does that sentence add?

Answer: A. No evidence; it is also generated text

Self-reassurance is not evidence.

✗ B. Proof that a person verified it first: nothing shows that

✗ C. A legal guarantee of accuracy: not a guarantee

✗ D. A link to the verification: no link was given

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F2-Q2 · question page 354

AI.LLM.04 · Check the claims that matter in a mixed answer

An answer gives four facts; you confirm three are right. What should you assume about the fourth?

Answer: D. Nothing yet; check it too if it matters

Mixed answers need claim-by-claim checks.

- ✗ A. It must be wrong since the others were right: unsupported
- ✗ B. It can be ignored without checking: only if it does not matter
- ✗ C. It must be right since the others were: errors can hide among correct facts

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F2-Q3 · question page 355

AI.LLM.04 · Check the claims that matter in a mixed answer

You are deciding whether to buy a used car. A model summary of the car's history is long and fluent. Which claims should you check first?

Answer: C. The claims the decision depends on, like accidents

Focus checks on decision-critical claims.

- ✗ A. None, since a long summary covers everything: length is not accuracy
- ✗ B. The sentence with the friendliest tone: tone does not set importance
- ✗ D. Only the first sentence, which sets the tone for the rest: not about importance

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F2-Q4 · question page 356

AI.LLM.04 · Check the claims that matter in a mixed answer

An answer includes a tidy reference list. What should you do before relying on it?

Answer: A. Open the references to see if they exist and agree

Generated references must be checked.

- ✗ B. Count the references; more means more accurate: count is not quality
- ✗ C. Trust it, since a reference list proves accuracy: lists can be generated
- ✗ D. Delete the list, since references are never real: some may be real

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S17-F2-Q5 · question page 357

AI.LLM.04 · Check the claims that matter in a mixed answer

Why can three correct facts make a fourth wrong fact more dangerous?

Answer: B. The correct facts make the wrong one seem trustworthy

Mixed accuracy lowers our guard.

- ✗ A. Readers always check the fourth fact first: unlikely
- ✗ C. Three correct facts make all facts correct: not true
- ✗ D. Wrong facts disappear when mixed with right ones: they do not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-M-Q1 · question page 359

AI.LLM.05 · Explain the context window and what happens when it is exceeded

What is a context window?

Answer: B. The limited amount of text a model can use at once

Context is the text the model can use right now.

- ✗ A. The time of day when the model is available: not this meaning
- ✗ C. The window on screen where the app is shown: not this meaning
- ✗ D. The list of every fact the model learned: that is training, not context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-M-Q2 · question page 360

AI.LLM.05 · Explain the context window and what happens when it is exceeded

Which of these is usually part of the context?

Answer: C. The conversation so far and any pasted text

Context holds the current conversation and supplied material.

- ✗ A. The model's entire training data: training data is not the context
- ✗ B. Every conversation all users have ever had: only this request's text
- ✗ D. The user's whole computer hard drive: only supplied text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-M-Q3 · question page 361

AI.LLM.05 · Explain the context window and what happens when it is exceeded

A very long conversation exceeds the window. What might happen, depending on the product?

Answer: A. Older messages may be dropped or shortened

Something must give when the limit is passed.

- ✗ B. The model becomes permanently smarter: not related
- ✗ C. The model rewrites its training data: not related
- ✗ D. All messages are kept forever with no limit: there is a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-M-Q4 · question page 362

AI.LLM.05 · Explain the context window and what happens when it is exceeded

Why might a model "forget" a detail from the start of a very long chat?

Answer: D. That detail may no longer be in its context

Text outside the context cannot be used.

✗ A. It chose to ignore the user on purpose: not intentional

✗ B. Models cannot read the first message ever: they can if it is in context

✗ C. Details from chats are erased from the internet: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-M-Q5 · question page 363

AI.LLM.05 · Explain the context window and what happens when it is exceeded

In what unit is a context window usually measured?

Answer: B. Tokens

Context limits are counted in tokens.

✗ A. Megapixels: an image measure

✗ C. Minutes: not a text measure

✗ D. Pages: not the usual unit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-B1-Q1 · question page 364

AI.LLM.05 · Compare the context window with a whiteboard

In the comparison, what is the whiteboard like?

Answer: D. The context window

The board holds what is in use now.

✗ A. The model's training data: training is not on the board

✗ B. The user's keyboard: not the comparison

✗ C. The company office: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-B1-Q2 · question page 365

AI.LLM.05 · Compare the context window with a whiteboard

What happens when the board fills up?

Answer: C. The oldest notes are erased

Old material is removed.

- ✗ A. All notes are kept: there is no room
- ✗ B. The board grows forever: it is not endless
- ✗ D. The meeting ends at once: not described

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-B1-Q3 · question page 366

AI.LLM.05 · Compare the context window with a whiteboard

Can the group use notes that were erased?

Answer: A. No, they are gone from the board

Erased text cannot be used.

- ✗ B. Yes, if they shout: not how it works
- ✗ C. Only on Fridays: nonsense
- ✗ D. Yes, the board remembers them: erased notes are gone

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-B1-Q4 · question page 367

AI.LLM.05 · Compare the context window with a whiteboard

What is written on the board?

Answer: B. Instructions, the chat and pasted text

The context holds current material.

- ✗ A. Only the newest single word: more than that

✗ C. Nothing at all: the board holds the discussion

✗ D. Everything ever said anywhere in the building: only this meeting

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-B1-Q5 · question page 368

AI.LLM.05 · Compare the context window with a whiteboard

Why might the group forget a point made at the start?

Answer: A. It may have been erased to make room

Early material can be removed.

✗ B. Points made first are always wrong: not the reason

✗ C. It was never said: it was said

✗ D. The board was too small to write on at all: it held the point at first

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F1-Q1 · question page 369

AI.LLM.05 · Work within a context window

After two hours of chat, a model ignores a formatting rule Kim gave at the start. What should she try?

Answer: C. Restate the rule in a new message

Restating puts the rule back into context.

✗ A. Assume the model is broken forever: a likely context limit

✗ B. Delete her account: unnecessary

✗ D. Wait until tomorrow: waiting does not restore context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F1-Q2 · question page 370

AI.LLM.05 · Work within a context window

Omar pastes a 400-page manual, and the answer seems to miss the last chapters. What is a likely reason?

Answer: D. Part of the manual may not fit in the context

Long inputs can exceed the context window.

- ✗ A. The last chapters were written in code: not implied
- ✗ B. The model already knew those chapters: not the issue
- ✗ C. Models refuse to read past page 100 by law: not a law

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F1-Q3 · question page 371

AI.LLM.05 · Work within a context window

Which approach helps with a very long document?

Answer: B. Split it into sections and do one at a time

Sections keep each request within limits.

- ✗ A. Paste the whole manual three times to be safe: uses even more context
- ✗ C. Remove all headings first: does not help
- ✗ D. Send it with no question at all: does not help

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F1-Q4 · question page 372

AI.LLM.05 · Work within a context window

A product says its context window is 8,000 tokens. Using about three quarters of a word per token, about how many words of total conversation can it consider?

Answer: A. About 6,000 words

8,000 x 0.75 = 6,000 words (an estimate).

✗ B. About 2,000 words: subtracted instead

✗ C. About 10,667 words: divided instead of multiplying

✗ D. Exactly 8,000 words: tokens are not words

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F1-Q5 · question page 373

AI.LLM.05 · Work within a context window

Why does starting a fresh conversation sometimes fix confused answers?

Answer: D. It clears old text that crowded the context

A new chat starts with a clean context.

✗ A. It deletes the model's parameters: not what happens

✗ B. It switches the user to a paid account: not implied

✗ C. It retrains the model from scratch: not what a new chat does

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F2-Q1 · question page 374

AI.LLM.05 · Help a model use long input well

A long report fits in the context, but the answer misses a key figure in the middle. What does this show?

Answer: C. Fitting in the window does not ensure careful use

Details in the middle can be overlooked.

✗ A. Models never read the middle of documents: too strong

✗ B. The figure must not exist in the report: it does

✗ D. Context windows are always unlimited: they are limited

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F2-Q2 · question page 375

AI.LLM.05 · Help a model use long input well

Where should you put the most important instructions in a long request?

Answer: B. Somewhere easy to find, then restate them

Prominent, repeated instructions help.

- ✗ A. Only in the file name: unlikely to be used
- ✗ C. Hidden in the middle among many other details: harder to use
- ✗ D. Nowhere; models guess instructions: they need them

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F2-Q3 · question page 376

AI.LLM.05 · Help a model use long input well

A tool advertises a much bigger context window. What can you conclude?

Answer: D. It takes more text; answers still need checking

A bigger limit is not a guarantee.

- ✗ A. It must use a completely different kind of model: not implied
- ✗ B. It now reads every detail perfectly: size is not careful reading
- ✗ C. It no longer makes any errors: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F2-Q4 · question page 377

AI.LLM.05 · Help a model use long input well

You need answers about sections 2 and 7 of a long contract. What is a sensible way to check the answer?

Answer: A. Confirm it covers both sections, using the text

Check coverage of the parts you care about.

- ✗ B. Check only section 2: misses section 7
- ✗ C. Accept it as long as the answer is long and detailed: length is not coverage
- ✗ D. Assume both were read: verify instead

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S18-F2-Q5 · question page 378

AI.LLM.05 · Help a model use long input well

A conversation has many similar product codes. The model mixes two up. What would help?

Answer: C. Restate the key codes clearly in this message

Clear, current restatement reduces mix-ups.

- ✗ A. Change the font of the codes: irrelevant
- ✗ B. Add even more similar codes: more confusion
- ✗ D. Ask the model to sound more confident about them: confidence is not accuracy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-M-Q1 · question page 380

AI.LLM.06 · Tell built-in knowledge from supplied or retrieved information

What is a training cutoff?

Answer: A. The date after which the training data has no information

Built-in knowledge stops at the cutoff.

- ✗ B. The time limit for typing a question: not this meaning
- ✗ C. The point where a model stops answering for the day: not this meaning
- ✗ D. The maximum number of users at once: not this meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-M-Q2 · question page 381

AI.LLM.06 · Tell built-in knowledge from supplied or retrieved information

A model's training cutoff was before a recent election. How could it still describe the result correctly?

Answer: B. If the result is supplied or retrieved for it

Current information must be supplied or retrieved.

- ✗ A. By training itself during the conversation: ordinary use does not retrain
- ✗ C. It cannot ever do so in any situation: supplied information makes it possible
- ✗ D. By guessing based on old patterns: a guess is not knowledge

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-M-Q3 · question page 382

AI.LLM.06 · Tell built-in knowledge from supplied or retrieved information

A product runs a web search and adds the results to the request. Which kind of information is that?

Answer: D. Retrieved information read now

Search results are retrieved into the context.

- ✗ A. Information that has no source at all: it comes from search results
- ✗ B. Built-in knowledge from training: it was fetched now
- ✗ C. A permanent change to the model: it does not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-M-Q4 · question page 383

AI.LLM.06 · Tell built-in knowledge from supplied or retrieved information

Which statement about retrieved information is accurate?

Answer: C. It is only as good as the source it came from

Retrieved text inherits its source's quality.

- ✗ A. It is the same as training data: it is read during the request
- ✗ B. It permanently updates the model's knowledge: it does not retrain
- ✗ D. It is always correct because it is recent: sources can be wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-M-Q5 · question page 384

AI.LLM.06 · Tell built-in knowledge from supplied or retrieved information

Which question most needs supplied or retrieved information rather than built-in knowledge?

Answer: D. What are this week's opening hours for the city pool?

Recent, changing facts need current sources.

- ✗ A. What is a triangle?: stable knowledge
- ✗ B. What is the boiling point of water at sea level?: stable knowledge
- ✗ C. What is a noun in English grammar?: stable knowledge

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-B1-Q1 · question page 385

AI.LLM.06 · Compare an encyclopedia on the shelf with today's newspaper

The encyclopedia is like which kind of information?

Answer: B. Built-in knowledge

Fixed at printing, like training.

- ✗ A. Retrieved information: that is the newspaper
- ✗ C. A web search: that would be retrieval

✗ D. Neither kind: it matches built-in knowledge

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-B1-Q2 · question page 386

AI.LLM.06 · Compare an encyclopedia on the shelf with today's newspaper

Today's newspaper is like which kind of information?

Answer: A. Supplied or retrieved text

Current material read now.

✗ B. The model's trained parameters: not stored there
✗ C. Training data: it is read now
✗ D. Built-in knowledge: that is the encyclopedia

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-B1-Q3 · question page 387

AI.LLM.06 · Compare an encyclopedia on the shelf with today's newspaper

Can the encyclopedia tell you yesterday's news?

Answer: C. No, it stops at its printing date

Built-in knowledge has a cutoff.

✗ A. Only on weekends: nonsense
✗ B. Yes, if you read it slowly: no
✗ D. Yes, encyclopedias update themselves: printed ones do not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-B1-Q4 · question page 388

AI.LLM.06 · Compare an encyclopedia on the shelf with today's newspaper

Does reading the newspaper change the encyclopedia?

Answer: B. No, it stays the same

Supplied text does not retrain the model.

- ✗ A. Yes, it rewrites the pages: no
- ✗ C. Yes, it adds a new volume: no
- ✗ D. Only if the news is big: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-B1-Q5 · question page 389

AI.LLM.06 · Compare an encyclopedia on the shelf with today's newspaper

Is the newspaper guaranteed to be correct?

Answer: D. No, it can contain mistakes

Retrieved sources can be wrong.

- ✗ A. Yes, because it is new: new is not correct
- ✗ B. Yes, all news is correct: not guaranteed
- ✗ C. Only if it is long: length is irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F1-Q1 · question page 390

AI.LLM.06 · Decide which kind of information an answer needs

Ben asks which phone model is newest. The tool does not search the web. What should he expect?

Answer: C. The answer may be out of date, reflecting its training cutoff

Without retrieval, answers reflect training-time knowledge.

- ✗ A. It will always name the newest model released today: it has a cutoff

✗ B. It will refuse to mention any phones: not implied

✗ D. It will search stores secretly: it does not search

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F1-Q2 · question page 391

AI.LLM.06 · Decide which kind of information an answer needs

A student pastes this year's club rules and asks about meeting times. What is the model using?

Answer: A. The supplied rules in the current request

Pasted text is supplied information.

✗ B. A web search of the school's site: nothing was searched

✗ C. Its memory of every club in the world: not built-in

✗ D. A retrained version of itself: pasting does not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F1-Q3 · question page 392

AI.LLM.06 · Decide which kind of information an answer needs

Which situation shows retrieval?

Answer: B. The product searches news sites and quotes them

Fetching and quoting sources is retrieval.

✗ A. The user types a short question with no files: no retrieval shown

✗ C. The model answers from what it learned in training: built-in knowledge

✗ D. The model writes a poem from imagination: no retrieval

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F1-Q4 · question page 393

AI.LLM.06 · Decide which kind of information an answer needs

A retrieved web page is three years old. What should you do before relying on it?

Answer: D. Check a current source for time-sensitive facts

Retrieved information can be outdated.

- ✗ A. Ask the model to make it newer: it cannot change the source
- ✗ B. Trust it, because it was retrieved by the product: old pages can be stale
- ✗ C. Assume nothing has changed: unsupported

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F1-Q5 · question page 394

AI.LLM.06 · Decide which kind of information an answer needs

Which question can a model usually answer from built-in knowledge without current sources?

Answer: C. What does the word "photosynthesis" mean?

Stable concepts suit built-in knowledge.

- ✗ A. Is the corner store open this holiday weekend?: needs current information
- ✗ B. Who won last night's game?: needs current information
- ✗ D. What is today's exchange rate?: needs current information

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F2-Q1 · question page 395

AI.LLM.06 · Untangle blended answers about recent facts

An answer quotes a supplied article correctly but adds a detail the article does not contain. Where might that detail have come from?

Answer: A. Older built-in knowledge or an invented guess

Blended answers can mix sources.

✗ B. The supplied article, which must contain it: it does not

✗ C. A law that requires extra details: nonsense

✗ D. A permanent update to the model: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F2-Q2 · question page 396

AI.LLM.06 · Untangle blended answers about recent facts

You want to know which parts of an answer came from your uploaded document. What is a useful request?

Answer: C. Ask it to mark statements taken from the document

Labeled sources make checking easier.

✗ A. Ask it to sound more confident about everything: confidence is not sourcing

✗ B. Ask it to stop using the document: defeats the purpose

✗ D. Ask for a longer answer: length is not sourcing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F2-Q3 · question page 397

AI.LLM.06 · Untangle blended answers about recent facts

A model describes "the latest version" of a product. How should you treat that phrase?

Answer: A. Check, since "latest" may reflect an old cutoff

"Latest" is time-sensitive.

✗ B. Trust it, since models always know the latest: cutoffs exist

✗ C. Assume no newer version can exist: unsupported

✗ D. Assume it is always wrong: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F2-Q4 · question page 398

AI.LLM.06 · Untangle blended answers about recent facts

A retrieved source is an unknown blog with no author or date. What is the best step?

Answer: D. Confirm it with a more reliable, dated source

Retrieved sources vary in quality.

- ✗ A. Trust it, since the product chose to retrieve it: retrieval is not endorsement
- ✗ B. Use it but remove the link: hides the problem
- ✗ C. Assume blogs are always wrong: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S19-F2-Q5 · question page 399

AI.LLM.06 · Untangle blended answers about recent facts

Which facts most need direct confirmation even after retrieval?

Answer: B. Prices, laws, schedules and officeholders

Time-sensitive facts change often.

- ✗ A. The meaning of common words: stable
- ✗ C. The plot of a famous novel from long ago: stable
- ✗ D. Basic arithmetic facts: stable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-M-Q1 · question page 401

AI.LLM.07 · Tell conversation context from saved memory

In one chat, you mention your dog's name, and later in the same chat the model uses it. What makes this possible?

Answer: A. The earlier message is still in the chat context

Earlier messages in the chat are part of the context.

- ✗ B. The model was retrained on your dog's name: no retraining happens
- ✗ C. The model guessed the name from training data: it read it from the chat
- ✗ D. Every model knows every pet's name: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-M-Q2 · question page 402

AI.LLM.07 · Tell conversation context from saved memory

A product greets you by name in a brand-new chat. What is the most likely reason?

Answer: D. The product saved a note and added it to the new chat

Saved memory is a product feature.

- ✗ A. The model's parameters changed to include your name: saved memory is not retraining
- ✗ B. The model read your mind from your typing: not possible
- ✗ C. All chatbots know every user's name: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-M-Q3 · question page 403

AI.LLM.07 · Tell conversation context from saved memory

Which statement about saved memory is accurate?

Answer: C. It is a product feature that varies and has settings

Memory varies by product and is usually controllable.

- ✗ A. It is part of every language model's design: it is a product feature
- ✗ B. It can never be viewed or deleted: often it can
- ✗ D. It permanently retrains the model for every user: it does not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-M-Q4 · question page 404

AI.LLM.07 · Tell conversation context from saved memory

Does conversation context retrain the model?

Answer: B. No; it is just text sent with each message

Context is supplied text, not training.

- ✗ A. Yes; every chat retrains the model: not during ordinary use
- ✗ C. Yes; but only on weekends: nonsense
- ✗ D. No; because context is deleted instantly: it is kept for the chat

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-M-Q5 · question page 405

AI.LLM.07 · Tell conversation context from saved memory

A friend says, "All chatbots remember everything forever." What is the best response?

Answer: C. It depends on the product; check its memory settings

Memory behavior is product-specific.

- ✗ A. True; models store chats in their parameters: not how it works
- ✗ B. False; no chatbot can ever save anything: some do
- ✗ D. True; every chatbot saves all chats forever: products differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-B1-Q1 · question page 406

AI.LLM.07 · Compare a conversation with a notebook

Saying "the movie I mentioned" during lunch is like what?

Answer: B. Conversation context

The current talk holds recent details.

- ✗ A. A web search: not described
- ✗ C. Saved memory: that is the notebook
- ✗ D. Training a new model: nothing is trained

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-B1-Q2 · question page 407

AI.LLM.07 · Compare a conversation with a notebook

The friend's notebook is like what?

Answer: A. Saved memory

Kept notes carry over to later talks.

- ✗ B. Conversation context: that is the lunch talk
- ✗ C. The model's parameters: not the comparison
- ✗ D. A search engine: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-B1-Q3 · question page 408

AI.LLM.07 · Compare a conversation with a notebook

Can the friend show you or erase the notebook?

Answer: D. Yes, it is kept separately and can be managed

Saved memory is often viewable and deletable.

- ✗ A. Only if it is full: not required
- ✗ B. No, it is secret forever: not described
- ✗ C. No, notebooks like that can never be erased or shown: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-B1-Q4 · question page 409

AI.LLM.07 · Compare a conversation with a notebook

Next month, how could the friend recall "likes horror movies"?

Answer: C. By reading the notebook

Saved notes carry over.

- ✗ A. By guessing: they wrote it down
- ✗ B. By asking a stranger: not described
- ✗ D. By remembering every word: the notebook keeps it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-B1-Q5 · question page 410

AI.LLM.07 · Compare a conversation with a notebook

Does keeping a notebook change who the friend is?

Answer: A. No, it is extra information they keep

Like memory, it does not retrain the model.

- ✗ B. Only if the notebook is long: no
- ✗ C. Yes, they become a new person entirely: no
- ✗ D. Yes, it rewires their brain: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F1-Q1 · question page 411

AI.LLM.07 · Manage context and memory in practice

Lia wants a product to stop bringing up her old projects in new chats. What should she check?

Answer: B. The product's memory or history settings

Memory settings control carried-over notes.

- ✗ A. Whether the model needs retraining: memory is a product feature
- ✗ C. Whether her internet is slow: unrelated
- ✗ D. Whether her messages are too short: unrelated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F1-Q2 · question page 412

AI.LLM.07 · Manage context and memory in practice

In a long chat, a model correctly refers to a recipe from 20 minutes earlier. What does it rely on?

Answer: D. The conversation context from this chat

Recent chat text is in context.

- ✗ A. Every recipe ever written: it used this chat
- ✗ B. A permanent change to its parameters: no retraining
- ✗ C. A separate saved note from last year: not needed here

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F1-Q3 · question page 413

AI.LLM.07 · Manage context and memory in practice

A product has no memory feature. You start a new chat. Should you expect it to know your earlier chat?

Answer: B. No; you would need to supply the details again

Without saved memory, new chats start fresh.

- ✗ A. No; and it will refuse to talk to you: it will talk, just without the history
- ✗ C. Yes; every model remembers every past chat: not without a memory feature

✗ D. Yes; the earlier chat retrained it: chats do not retrain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F1-Q4 · question page 414

AI.LLM.07 · Manage context and memory in practice

Which action is about saved memory rather than conversation context?

Answer: A. Viewing and deleting stored notes in settings

Stored notes are product memory.

✗ B. Scrolling up to reread this chat: context
✗ C. Pasting a file into this message: supplied context
✗ D. Restating a rule in the current chat window: context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F1-Q5 · question page 415

AI.LLM.07 · Manage context and memory in practice

A coworker says, "I told it my preference once, so the model learned it forever." What is more accurate?

Answer: C. A product may have saved a note; the model did not retrain

Saved notes are separate from the model.

✗ A. Correct; all preferences are added to its training right away: not automatically
✗ B. Wrong; products never save preferences: some do
✗ D. Correct; models retrain after each preference: they do not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F2-Q1 · question page 416

AI.LLM.07 · Handle memory features carefully

A family shares one account, and a new answer mentions a private detail one member shared. What is the issue?

Answer: D. Saved memory on a shared account can expose details

Shared accounts plus memory can leak details.

- ✗ A. The model hacked the family's computer: not implied
- ✗ B. Memory features never store private details: they may
- ✗ C. This is impossible with any product: possible with saved memory

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F2-Q2 · question page 417

AI.LLM.07 · Handle memory features carefully

Jay turns memory off. Does that automatically delete chats already stored?

Answer: B. Not necessarily; read what the setting does

Check what each setting actually does.

- ✗ A. No, chats can never be deleted by anyone: often they can
- ✗ C. Yes, always, in every product: products differ
- ✗ D. Yes, and it retrains the model: no retraining

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F2-Q3 · question page 418

AI.LLM.07 · Handle memory features carefully

Which is the safest habit for very sensitive details, such as medical history?

Answer: D. Do not share details you would not want stored

Not sharing is the most reliable protection.

✗ A. Share them with a friendly greeting: tone does not protect

✗ B. Share them, since settings always protect you: settings vary

✗ C. Share them only in long conversations: length does not protect

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F2-Q4 · question page 419

AI.LLM.07 · Handle memory features carefully

A user deletes a chat but later sees a detail from it in a new answer. What is a possible explanation?

Answer: C. The product saved that detail separately as a note

Saved notes can be separate from chat history.

✗ A. The new chat is the same chat: it is new

✗ B. The detail was in the model's original training: unlikely for a personal detail

✗ D. Deleting chats retrains the model: not how it works

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S20-F2-Q5 · question page 420

AI.LLM.07 · Handle memory features carefully

What is the best source for how a specific product handles memory?

Answer: A. Its own settings and help pages

Product documentation is the best evidence.

✗ B. A general article about all chatbots: too general

✗ C. A friend's guess: hearsay

✗ D. The model's answer about itself: may be inaccurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-M-Q1 · question page 422

AI.LLM.09 · Explain why a product's behavior is more than its model

What are system instructions?

Answer: B. Directions the company adds to shape answers

Companies write instructions that guide their product.

- ✗ A. Questions the user types into the chat box: those are user messages
- ✗ C. The operating system of your phone: not this meaning
- ✗ D. The training data used to build the model: not instructions

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-M-Q2 · question page 423

AI.LLM.09 · Explain why a product's behavior is more than its model

Two products use similar models, but one refuses a request the other accepts. What is a likely reason?

Answer: A. Different instructions or filters in each product

Product layers shape refusals.

- ✗ B. Similar models always behave identically in every product: products differ
- ✗ C. The user typed the request too slowly: irrelevant
- ✗ D. One model must be completely broken: usually a product difference

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-M-Q3 · question page 424

AI.LLM.09 · Explain why a product's behavior is more than its model

A product can read your spreadsheet files. Where does that ability come from?

Answer: D. A file-reading tool the product provides

Tools are added by the product.

- ✗ A. The model memorized your spreadsheet in training: not possible
- ✗ B. Every language model reads files by itself: tools are product features
- ✗ C. Your spreadsheet program sends answers automatically: not how it works

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-M-Q4 · question page 425

AI.LLM.09 · Explain why a product's behavior is more than its model

What is the best way to learn how a specific product behaves?

Answer: C. Test it on your task and read its documentation

Product-specific testing and docs are best.

- ✗ A. Read a general description of all language models: too general
- ✗ B. Ask a friend who uses a different product: different product
- ✗ D. Assume it matches the first chatbot you used: products differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-M-Q5 · question page 426

AI.LLM.09 · Explain why a product's behavior is more than its model

Which is an example of a product limit?

Answer: A. A maximum length for each answer

Products set limits on length and usage.

- ✗ B. The meaning of a word: not a limit
- ✗ C. The user's choice of topic: not a product limit
- ✗ D. The patterns learned in training: model, not limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-B1-Q1 · question page 427

AI.LLM.09 · Compare a restaurant's kitchen with its menu

In the comparison, the model is like what?

Answer: C. The supplier's ingredients

The model is the shared ingredient.

- ✗ A. The pizza oven: that is like a tool
- ✗ B. The house rules: those are like instructions
- ✗ D. The portion sizes it serves: those are like limits

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-B1-Q2 · question page 428

AI.LLM.09 · Compare a restaurant's kitchen with its menu

A "no substitutions" rule is like which product layer?

Answer: B. System instructions or filters

House rules shape what is allowed.

- ✗ A. The model's original training: not the comparison
- ✗ C. The supplier: that is the model
- ✗ D. The user's order: not a house rule

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-B1-Q3 · question page 429

AI.LLM.09 · Compare a restaurant's kitchen with its menu

One restaurant has a pizza oven. In AI terms, that is like what?

Answer: D. A tool the product provides

Equipment enables extra abilities, like tools.

- ✗ A. A safety filter: a filter blocks, a tool enables
- ✗ B. The model's learned parameters: not equipment
- ✗ C. Training data: not equipment

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-B1-Q4 · question page 430

AI.LLM.09 · Compare a restaurant's kitchen with its menu

Why can two restaurants using the same supplier serve different meals?

Answer: B. Their recipes, rules, equipment and portions differ

Layers on top create differences.

- ✗ A. The supplier changes ingredients for each one: same supplier
- ✗ C. Customers cook the meals themselves: not described
- ✗ D. They cannot; same supplier means same meals: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-B1-Q5 · question page 431

AI.LLM.09 · Compare a restaurant's kitchen with its menu

Knowing the supplier tells you how much about the meal?

Answer: D. Only a little

Like knowing the model: helpful but incomplete.

- ✗ A. Everything about it: layers matter
- ✗ B. Nothing at all: it tells you some
- ✗ C. Exactly the price: not determined

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F1-Q1 · question page 432

AI.LLM.09 · Predict and check product behavior

A school's chatbot refuses to write complete essays, but a public chatbot will. What explains this best?

Answer: A. The school product added limits on essay writing

Product instructions differ.

- ✗ B. Essays cannot be generated by any model: the public one did
- ✗ C. The school chatbot cannot write sentences at all: it can
- ✗ D. Public chatbots ignore all instructions: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F1-Q2 · question page 433

AI.LLM.09 · Predict and check product behavior

An article says "language models cannot browse the web." Your product shows web results. What is true?

Answer: C. Your product adds a browsing tool to its model

General statements about models may not describe products.

- ✗ A. Browsing is part of every model: it is a product tool
- ✗ B. The article proves your product is fake: products add tools
- ✗ D. The model learned the web results in training: they were fetched now

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F1-Q3 · question page 434

AI.LLM.09 · Predict and check product behavior

Before relying on a product to summarize contracts, what should a manager do?

Answer: B. Test it on sample contracts and read its documentation

Test the specific product on the task.

- ✗ A. Trust a general article about language models: too general
- ✗ C. Assume all products summarize equally well: they differ
- ✗ D. Ask the product if it is good at contracts: not a test

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F1-Q4 · question page 435

AI.LLM.09 · Predict and check product behavior

A product gives very short answers no matter what you ask. What is a likely reason?

Answer: C. Its instructions or settings limit answer length

Product settings shape length.

- ✗ A. All language models only give short answers: not true
- ✗ B. Your device has a broken keyboard: unrelated
- ✗ D. The model has run out of words forever: not how it works

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F1-Q5 · question page 436

AI.LLM.09 · Predict and check product behavior

Which statement is most accurate?

Answer: A. Products built on similar models can behave quite differently

Product layers matter.

- ✗ B. Products never change how models behave: they do
- ✗ C. A model description predicts product behavior exactly: it does not

✗ D. Any two products with similar models act the same: layers differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F2-Q1 · question page 437

AI.LLM.09 · Plan for product changes

A workflow that summarized meeting notes worked last month but now formats answers differently. What is a likely explanation?

Answer: D. The product updated its instructions or model

Products change over time.

✗ A. Your computer's clock is wrong: unrelated
✗ B. The workflow was never possible: it worked before
✗ C. The meeting notes became illegal to summarize: unlikely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F2-Q2 · question page 438

AI.LLM.09 · Plan for product changes

Where is the best record of what changed in a product?

Answer: A. Its release notes and help pages

Official documentation records changes.

✗ B. A guess from a friend: not a record
✗ C. An old screenshot from last year: may be outdated
✗ D. The model's description of itself: may be inaccurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F2-Q3 · question page 439

AI.LLM.09 · Plan for product changes

A team relies on a product for weekly reports. What habit protects them from surprises?

Answer: D. Keep test cases and rerun them after updates

Retesting catches changes.

- ✗ A. Switch products every week: disruptive
- ✗ B. Never look at the outputs again once it works: risky
- ✗ C. Assume updates never affect results: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F2-Q4 · question page 440

AI.LLM.09 · Plan for product changes

You noticed once that a product ignores a filter if you phrase things oddly. Should you build a process on this?

Answer: B. No; undocumented behavior can stop at any time

Do not rely on undocumented behavior.

- ✗ A. No; because products never change: they do change
- ✗ C. Yes; odd phrasing is an official feature of the product: not documented
- ✗ D. Yes; once noticed, it will always work: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S21-F2-Q5 · question page 441

AI.LLM.09 · Plan for product changes

A limit that blocked a request last month is gone. What is the most accurate conclusion?

Answer: C. The product changed its limits; check its documentation

Limits are product decisions that change.

- ✗ A. The request must now be unsafe: unsupported

✗ B. Limits never existed: it existed before

✗ D. The model learned on its own to ignore its limits: models do not change limits

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-M-Q1 · question page 443

AI.PROMPT.01 · Identify the task, context, audience and purpose a request needs

In "Explain (the action word) photosynthesis (the process) to a group of eight-year-olds," which part names the audience?

Answer: D. a group of eight-year-olds

The audience is who will read or use it.

✗ A. to: only a connecting word

✗ B. photosynthesis (the process): that is the topic

✗ C. Explain (the action word): that is the task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-M-Q2 · question page 444

AI.PROMPT.01 · Identify the task, context, audience and purpose a request needs

"Write about recycling." Which pieces are missing?

Answer: B. Context, audience and purpose

The task is there; context, audience and purpose are not.

✗ A. Only the task, which is not stated at all: the task "write" is stated

✗ C. Nothing is missing; the request is complete: three pieces are missing

✗ D. Only the topic of the writing: the topic is recycling

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-M-Q3 · question page 445

AI.PROMPT.01 · Identify the task, context, audience and purpose a request needs

Which detail is CONTEXT the tool could not know on its own?

Answer: A. "Our school recycles paper, not plastic."

Facts about your situation are context.

- ✗ B. "Please answer in English.": a language instruction
- ✗ C. "Make it sound friendly and clear.": a style instruction
- ✗ D. "Use short sentences in the reply.": a format instruction

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-M-Q4 · question page 446

AI.PROMPT.01 · Identify the task, context, audience and purpose a request needs

Which phrase states a PURPOSE?

Answer: C. "so the class votes for a compost bin"

Purpose says why the result is needed.

- ✗ A. "about 200 words long": that is a length limit
- ✗ B. "for the class newsletter next week": that is where it appears
- ✗ D. "write a short article": that is the task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-M-Q5 · question page 447

AI.PROMPT.01 · Identify the task, context, audience and purpose a request needs

Why does a vague request often get a generic answer?

Answer: D. The tool has to guess the missing pieces

Missing details lead to guesses.

✗ A. Short requests use too many tokens: the opposite

✗ B. Vague requests are always rejected by filters: not true

✗ C. Tools refuse to answer short requests: they answer, generically

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-B1-Q1 · question page 448

AI.PROMPT.01 · Separate a request from the background it needs

In "make dinner," what is missing?

Answer: A. The background, such as allergies

The background is missing.

✗ B. The house address: they are already there

✗ C. The request itself: "make dinner" is the request

✗ D. The babysitter's full name and age: not needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-B1-Q2 · question page 449

AI.PROMPT.01 · Separate a request from the background it needs

"One child is allergic to nuts" is which part?

Answer: C. Background (context)

It is information the babysitter cannot see.

✗ A. A greeting: not a greeting

✗ B. The request: it does not say what to do

✗ D. The audience for dinner: not who reads it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-B1-Q3 · question page 450

AI.PROMPT.01 · Separate a request from the background it needs

Why might the request feel complete to the parent?

Answer: B. The parent already knows the background

Known background is easy to forget to say.

- ✗ A. Babysitters can read minds: they cannot
- ✗ C. Requests never need details: they often do
- ✗ D. The kids will explain everything to the babysitter: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-B1-Q4 · question page 451

AI.PROMPT.01 · Separate a request from the background it needs

What could happen without the background?

Answer: A. An unsafe or unwanted dinner

Missing context leads to bad guesses.

- ✗ B. The kids cooking instead: not described
- ✗ C. A perfect dinner every time: unlikely without key facts
- ✗ D. No dinner at all, ever: too extreme

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-B1-Q5 · question page 452

AI.PROMPT.01 · Separate a request from the background it needs

In an AI request, what does context provide?

Answer: B. Facts the tool cannot see for itself

Context fills in your situation.

- ✗ A. The tool's price: not context

✗ C. The tool's training data: not context

✗ D. A list of the tool's display settings: not context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F1-Q1 · question page 453

AI.PROMPT.01 · Choose an improved request for a stated need

Pat needs a short email asking her landlord to fix a leaking sink before Friday. Which request is best?

Answer: C. "Draft a polite email asking my landlord to fix the sink by Friday."

It states task, audience and purpose.

✗ A. "Explain how sinks leak and how plumbers fix them.": not her task

✗ B. "Write something about landlords in general.": too vague

✗ D. "Write an email to my landlord about things in my apartment.": no task details, audience or purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F1-Q2 · question page 454

AI.PROMPT.01 · Choose an improved request for a stated need

A student wants a study guide for a biology test on cells. What is the most useful detail to add to "Make a study guide"?

Answer: D. The topic, the course level and the test format

Topic, level and format are key context.

✗ A. The color of the student's notebook: irrelevant

✗ B. A request to make the guide sound impressive: does not help it fit

✗ C. The name of the student's best friend: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F1-Q3 · question page 455

AI.PROMPT.01 · Choose an improved request for a stated need

Which request names a clear audience?

Answer: C. "Explain credit scores for a high-school senior."

A specific reader shapes the explanation.

- ✗ A. "Explain credit scores in some detail for me.": audience is vague
- ✗ B. "Explain credit scores very well, please.": no audience
- ✗ D. "Explain credit scores, and make it good.": no audience

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F1-Q4 · question page 456

AI.PROMPT.01 · Choose an improved request for a stated need

Marco asks, "Give me ideas." What should he add first?

Answer: D. What the ideas are for and any limits

Purpose and limits make ideas useful.

- ✗ A. A request for the ideas to be numbered in bold: formatting before purpose
- ✗ B. Nothing; short requests always work best: too vague to help
- ✗ C. The word "please" at the start of the sentence: politeness is not context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F1-Q5 · question page 457

AI.PROMPT.01 · Choose an improved request for a stated need

Which request includes context the tool could not guess?

Answer: A. "Plan meals for a week; we have no oven, only a stove."

Having no oven is key context.

- ✗ B. "Plan a week of meals as fast as you can.": no situational context
- ✗ C. "Plan a week of meals, please and thank you.": no situational context
- ✗ D. "Plan meals for a week that taste really good.": no situational context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F2-Q1 · question page 458

AI.PROMPT.01 · Find the missing or conflicting piece before sending

"Explain this contract simply for my grandmother, in full legal detail." What is the problem?

Answer: B. It asks for two different audiences at once

Simple-for-a-grandmother and full legal detail conflict.

- ✗ A. Grandmothers cannot read contracts: not true
- ✗ C. It is too short to be understood at all: length is not the issue
- ✗ D. It does not say what the contract is about: the main issue is the conflict

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F2-Q2 · question page 459

AI.PROMPT.01 · Find the missing or conflicting piece before sending

"Fix the schedule." What does this request assume?

Answer: A. That the tool knows which schedule and what is wrong

Key details are assumed, not stated.

- ✗ B. That the tool can see every calendar in the world: too extreme a reading
- ✗ C. That schedules never need fixing at all: not implied
- ✗ D. That the user wants a poem: not implied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F2-Q3 · question page 460

AI.PROMPT.01 · Find the missing or conflicting piece before sending

What is a good test before sending an important request?

Answer: B. Reread it as a stranger with no background would

Strangers reveal missing pieces.

- ✗ A. Count the words and make sure it is very short: length is not the test
- ✗ C. Send it and hope for the best: skips the check
- ✗ D. Add the word "urgent" to the start: urgency is not clarity

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F2-Q4 · question page 461

AI.PROMPT.01 · Find the missing or conflicting piece before sending

"Summarize the report for the board so they can decide on the budget." Which piece is still missing?

Answer: C. Which report (it must be supplied)

The report itself is missing context.

- ✗ A. The purpose, which is not mentioned anywhere: deciding the budget is the purpose
- ✗ B. The task, which is not mentioned anywhere: summarize is the task
- ✗ D. The audience, which is not mentioned anywhere: the board is the audience

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S22-F2-Q5 · question page 462

AI.PROMPT.01 · Find the missing or conflicting piece before sending

A request asks for "a fun, serious speech." What should the writer do first?

Answer: D. Decide which tone matters more, or how to balance them

Resolve conflicting requirements first.

- ✗ A. Send it as written; tools resolve conflicts well: conflicts lead to muddled results
- ✗ B. Ask for ten speeches at once: does not resolve the conflict
- ✗ C. Remove the word "speech" from the request: removes the task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-M-Q1 · question page 464

AI.PROMPT.02 · State the output format and constraints a result must meet

Which part of a request describes the OUTPUT FORMAT?

Answer: B. "as a table with Item and Price columns"

Format is the shape of the result.

- ✗ A. "compare three caterers": that is the task
- ✗ C. "for my coworkers": that is the audience
- ✗ D. "so we can choose a caterer by Friday": that is the purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-M-Q2 · question page 465

AI.PROMPT.02 · State the output format and constraints a result must meet

Which is a CONSTRAINT?

Answer: D. "under 150 words"

A length limit is a constraint.

- ✗ A. "write a cover letter": that is the task
- ✗ B. "for a hiring manager": that is the audience
- ✗ C. "to apply for a job": that is the purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-M-Q3 · question page 466

AI.PROMPT.02 · State the output format and constraints a result must meet

In this fictional example, which instruction did the response NOT follow?

Fictional example (prepared in advance; not a real AI output)

Request:

> List three tips for saving money on groceries. Use a numbered list. Keep each tip under 15 words.

Response:

> 1. Plan your meals for the week before you go shopping.
> 2. Compare unit prices instead of package prices on the shelf.
> 3. Buy store brands, which are often cheaper than name brands, especially for basic items like flour, sugar, rice and canned vegetables.

Answer: C. Keeping each tip under 15 words

Tip 3 is far longer than 15 words.

- ✗ A. Giving three tips: it gave three
- ✗ B. Using a numbered list: it used one
- ✗ D. Keeping the topic on groceries: it did

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-M-Q4 · question page 467

AI.PROMPT.02 · State the output format and constraints a result must meet

Why does stating format up front help?

Answer: A. It saves revisions and makes checking easy

A clear shape is quick to verify.

- ✗ B. It makes the tool respond more slowly: not a benefit
- ✗ C. It guarantees the facts are correct: format does not check facts
- ✗ D. It lets you skip reading the result: you still need to read it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-M-Q5 · question page 468

AI.PROMPT.02 · State the output format and constraints a result must meet

Which instruction is something to AVOID, a kind of constraint?

Answer: B. "Do not mention prices."

An "avoid" rule is a constraint.

- ✗ A. "Explain the field trip.": task
- ✗ C. "Write it for parents.": audience
- ✗ D. "Make a checklist.": format

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-B1-Q1 · question page 469

AI.PROMPT.02 · Turn a vague wish into a checkable instruction

Which instruction is checkable?

Answer: D. "Under 100 words"

You can count to check it.

- ✗ A. "Not too long": vague
- ✗ B. "Brief-ish": vague
- ✗ C. "Short": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-B1-Q2 · question page 470

AI.PROMPT.02 · Turn a vague wish into a checkable instruction

Which is the most checkable version of "nice looking"?

Answer: A. "A table with three columns"

Columns can be counted.

- ✗ B. "Stylish": vague
- ✗ C. "Pretty and neat": vague
- ✗ D. "Very professional-looking": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-B1-Q3 · question page 471

AI.PROMPT.02 · Turn a vague wish into a checkable instruction

What question helps turn a wish into a checkable instruction?

Answer: C. "How would I know it was done?"

Define what success looks like.

- ✗ A. "How do I sound polite?": not about checking
- ✗ B. "Which tool is cheapest this month?": not about checking
- ✗ D. "How fast can it answer?": not about checking

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-B1-Q4 · question page 472

AI.PROMPT.02 · Turn a vague wish into a checkable instruction

Which instruction is easier to test than "friendly"?

Answer: A. "No technical terms"

You can scan for technical terms.

✗ B. "Happy-sounding": vague

✗ C. "Warm": vague

✗ D. "Kind and fun": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-B1-Q5 · question page 473

AI.PROMPT.02 · Turn a vague wish into a checkable instruction

Which result meets "exactly five bullet points"?

Answer: C. A list with five bullets

Five bullets meets the instruction.

✗ A. A paragraph with no bullets: no bullets

✗ B. A list with four bullets: one short

✗ D. A list with seven bullets: two too many

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F1-Q1 · question page 474

AI.PROMPT.02 · Write format and constraints for real requests

Lee needs a sign-up sheet for a car wash with columns for name, phone and time slot. Which instruction fits?

Answer: D. "Make a table with columns Name, Phone and Time slot."

Named columns give a checkable format.

✗ A. "Make it look really nice and clean.": vague

✗ B. "Make something people can sign up on, please.": format not specified

✗ C. "Write a story about a car wash.": wrong task

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F1-Q2 · question page 475

AI.PROMPT.02 · Write format and constraints for real requests

A form allows only 280 characters. Which constraint should Ana state?

Answer: B. "Under 280 characters, including spaces."

State the exact limit.

- ✗ A. "Use as many words as you need.": ignores the limit
- ✗ C. "Keep it reasonably short and sweet.": not checkable
- ✗ D. "Make it sound exciting.": a tone, not the limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F1-Q3 · question page 476

AI.PROMPT.02 · Write format and constraints for real requests

A coach wants practice drills a beginner can follow. Which constraint fits best?

Answer: C. "Numbered steps, no jargon, for beginners."

Format, vocabulary and level match the need.

- ✗ A. "Mention every famous athlete you know.": irrelevant
- ✗ B. "Any format is fine, whatever you like.": misses the need
- ✗ D. "Make it as advanced as possible.": wrong level

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F1-Q4 · question page 477

AI.PROMPT.02 · Write format and constraints for real requests

Which request would produce a result that is easiest to check?

Answer: D. "List exactly five tips, each under 20 words."

Exact counts and limits are easy to verify.

✗ A. "Some tips, please, as good as you can make them.": vague

✗ B. "Give me some tips, not too many.": vague count

✗ C. "Write a lot of helpful tips.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F1-Q5 · question page 478

AI.PROMPT.02 · Write format and constraints for real requests

A teacher wants a parent letter without mentioning grades. How should she say this?

Answer: A. "Do not mention grades."

A clear "avoid" rule is easy to follow and check.

✗ B. "Make it nice for parents.": does not state the rule

✗ C. "Use good judgment on content.": too vague

✗ D. "Be careful about some topics.": too vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F2-Q1 · question page 479

AI.PROMPT.02 · Check whether constraints were actually followed

A response says, "Here is your summary in under 100 words," but it looks long. What should you do?

Answer: B. Count the words yourself

Verify constraints that matter.

✗ A. Trust the claim, since it said it complied: claims are not checks

✗ C. Ask it to repeat the claim: still not a check

✗ D. Assume all summaries are under 100 words: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F2-Q2 · question page 480

AI.PROMPT.02 · Check whether constraints were actually followed

A request lists twelve formatting rules, and the result breaks two. What is a sensible next step?

Answer: D. Point out the two broken rules and ask for a fix

Name specific failures and recheck.

- ✗ A. Add twelve more rules: more rules can make it worse
- ✗ B. Accept it without checking the rest: other rules may also fail
- ✗ C. Give up on using any formatting rules from now on: too extreme

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F2-Q3 · question page 481

AI.PROMPT.02 · Check whether constraints were actually followed

Which version of "keep it short" can be checked?

Answer: B. "Three sentences or fewer"

A sentence count can be checked.

- ✗ A. "Short and snappy, please": vague
- ✗ C. "Brief, as usual": vague
- ✗ D. "Not wordy": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F2-Q4 · question page 482

AI.PROMPT.02 · Check whether constraints were actually followed

A grant form requires exactly 500 characters or fewer. The tool says its text is 480 characters. What is the safest action?

Answer: A. Check the character count before submitting

Verify exact limits yourself.

- ✗ B. Ask the tool if it is sure: not a real check
- ✗ C. Make it longer to be safe: risks breaking the limit
- ✗ D. Submit it, since the tool stated the count: tools can miscount

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S23-F2-Q5 · question page 483

AI.PROMPT.02 · Check whether constraints were actually followed

Why might a tool miss an exact word count even when asked clearly?

Answer: C. Exact counts are hard for it to track precisely

Precise counting is a known weak spot; check it.

- ✗ A. Word counts are illegal to request: nonsense
- ✗ B. Tools are designed to ignore every limit on purpose: not true
- ✗ D. Exact counts only work in other languages: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-M-Q1 · question page 485

AI.PROMPT.03 · Use examples to show what you want, and notice when they pull too hard

Why include a short example of the desired output?

Answer: A. It shows the style, length and structure to copy

Examples demonstrate the pattern.

- ✗ B. It makes the reply arrive faster than usual: not the purpose
- ✗ C. It proves the facts in the answer are true: examples do not verify facts
- ✗ D. It makes the tool ignore all other instructions: not the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-M-Q2 · question page 486

AI.PROMPT.03 · Use examples to show what you want, and notice when they pull too hard

In this fictional example, what went wrong?

Fictional example (prepared in advance; not a real AI output)

Request:

Write three quiz questions about planets in the style of this example.
Example: "Which planet is known as the Red Planet? (Mars)"

Response:

1. Which planet is known as the Red Planet? (Mars)
2. Which planet is often called the Red Planet? (Mars)
3. Which planet has a reddish color from iron oxide? (Mars)

Answer: C. Every question copied the example's topic, Mars

The tool copied the example too closely.

✗ A. The questions did not use the example's format: they did

✗ B. The questions were about the wrong subject entirely: they were about planets

✗ D. There were too many questions: there were three

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-M-Q3 · question page 487

AI.PROMPT.03 · Use examples to show what you want, and notice when they pull too hard

What would most likely fix the problem in that example?

Fictional example (prepared in advance; not a real AI output)

Request:

Write three quiz questions about planets in the style of this example.
Example: "Which planet is known as the Red Planet? (Mars)"

Response:

1. Which planet is known as the Red Planet? (Mars)
2. Which planet is often called the Red Planet? (Mars)
3. Which planet has a reddish color from iron oxide? (Mars)

Answer: B. Saying "use different planets than the example"

Say what should differ from the example.

✗ A. Adding a second example that is also about Mars: reinforces the copying

✗ C. Removing the word "quiz" from the request: not the problem

✗ D. Asking for thirty questions instead of three: more copies likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-M-Q4 · question page 488

AI.PROMPT.03 · Use examples to show what you want, and notice when they pull too hard

How should an example be marked in a request?

Answer: D. Clearly labeled, like "Example of my style:"

Labels separate examples from instructions.

✗ A. Written in capital letters only: not a clear label

✗ B. Hidden in the middle of the request with no label: may be confused with instructions

✗ C. Placed after the reply is finished: too late

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-M-Q5 · question page 489

AI.PROMPT.03 · Use examples to show what you want, and notice when they pull too hard

What is a risk of including examples?

Answer: C. The tool may copy them too closely

Examples can over-steer the output.

- ✗ A. The tool will refuse to answer: not a typical risk
- ✗ B. The answer will be in another language: not a typical risk
- ✗ D. The tool will always ignore them: often the opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-B1-Q1 · question page 490

AI.PROMPT.03 · Compare examples with a sample a tailor copies

What does the tailor copy from the sample shirt?

Answer: A. Its fit, collar and length

Samples show many features.

- ✗ B. Only the store it came from: not described
- ✗ C. Only its price tag: not described
- ✗ D. Nothing at all: the tailor copies it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-B1-Q2 · question page 491

AI.PROMPT.03 · Compare examples with a sample a tailor copies

Why might you get the same color?

Answer: B. You did not say the color should differ

Unstated differences get copied.

- ✗ A. The tailor has only one color of cloth: not stated

✗ C. The shirt was the wrong size: unrelated

✗ D. Colors cannot be changed: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-B1-Q3 · question page 492

AI.PROMPT.03 · Compare examples with a sample a tailor copies

What tends to win: the shirt in hand or a quick comment?

Answer: D. The shirt in hand

Concrete samples are powerful.

✗ A. Neither: the sample wins

✗ B. The tailor's mood: not described

✗ C. The quick comment: the sample tends to dominate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-B1-Q4 · question page 493

AI.PROMPT.03 · Compare examples with a sample a tailor copies

In a request, what is like the sample shirt?

Answer: C. An example of the output

Examples are samples.

✗ A. The user's name: not the comparison

✗ B. The tool's privacy settings: not the comparison

✗ D. The tool's price: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-B1-Q5 · question page 494

AI.PROMPT.03 · Compare examples with a sample a tailor copies

How do you get a different color from the tailor?

Answer: B. Say clearly what should differ

State the differences.

- ✗ A. Ask for the same shirt twice: repeats the problem
- ✗ C. Say nothing: the color gets copied
- ✗ D. Bring the same shirt again: repeats the problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F1-Q1 · question page 495

AI.PROMPT.03 · Write requests with helpful examples

Dana wants product descriptions like her best one, but for new products. What should she write?

Answer: D. Her best one, labeled, plus "use the new details"

A labeled example plus what should differ.

- ✗ A. Her best description pasted with no label or note: may be copied too closely
- ✗ B. A list of products with no example at all: loses the style she wants
- ✗ C. Only "write some product descriptions": no example or details

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F1-Q2 · question page 496

AI.PROMPT.03 · Write requests with helpful examples

A teacher gives one sample math word problem about apples. The results all mention apples. What should she change?

Answer: A. Add a varied second example or ask for other topics

Varied examples or explicit differences help.

✗ B. Nothing; apples are required in math problems: not true

✗ C. Give five more apple problems as examples: reinforces apples

✗ D. Remove all instructions and keep the example: makes copying worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F1-Q3 · question page 497

AI.PROMPT.03 · Write requests with helpful examples

Which request best uses an example?

Answer: C. "Style example: 'Bright, simple.' Write three for new products."

A labeled example and a clear task.

✗ A. "Copy this exactly: 'Bright, simple.'": asks to copy, not to follow a style

✗ B. "Write three descriptions for new products, nicely.": no example given

✗ D. "Write three product descriptions in a style I would like.": no example given

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F1-Q4 · question page 498

AI.PROMPT.03 · Write requests with helpful examples

Why label an example clearly?

Answer: A. So it is not mistaken for an instruction

Labels separate examples from the request.

✗ B. So the reply becomes longer: not the purpose

✗ C. So the tool can skip reading the example: it should read it

✗ D. So the facts become verified: labels do not verify

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F1-Q5 · question page 499

AI.PROMPT.03 · Write requests with helpful examples

An example email is formal, but you want a casual email. What is the simplest fix?

Answer: D. Use a casual example, or describe the tone

Examples should match what you want.

- ✗ A. Keep the formal example and say nothing else: tone will likely be copied
- ✗ B. Ask for "an email" with no example: loses guidance
- ✗ C. Add three more formal examples: reinforces formality

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F2-Q1 · question page 500

AI.PROMPT.03 · Fix conflicts between examples and instructions

Instructions say "four sentences each," but the only example has two sentences. What result is likely?

Answer: B. Short descriptions that follow the example

Examples tend to override conflicting rules.

- ✗ A. Descriptions in a different language: not likely
- ✗ C. No descriptions at all: not likely
- ✗ D. Exactly four sentences every time: the example often wins

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F2-Q2 · question page 501

AI.PROMPT.03 · Fix conflicts between examples and instructions

Every example mentions "Brand X." The new descriptions keep mentioning Brand X. What is the best fix?

Answer: A. Use examples without Brand X or vary them

Remove the repeated detail from examples.

- ✗ B. Add more examples with Brand X: makes it worse
- ✗ C. Remove the instructions but keep the examples: makes it worse
- ✗ D. Ask for longer descriptions: does not fix the copying

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F2-Q3 · question page 502

AI.PROMPT.03 · Fix conflicts between examples and instructions

After fixing a conflicting example, what should you still check?

Answer: C. That results follow the written rules

Check outputs against the rules.

- ✗ A. Nothing; fixing the example guarantees success: still check
- ✗ B. Only that results match the example's topic: that was the problem
- ✗ D. Only the number of words in the request: not the result

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S24-F2-Q4 · question page 503

AI.PROMPT.03 · Fix conflicts between examples and instructions

When might it be better to remove an example and describe the pattern in words?

Answer: D. When the example keeps pulling results the wrong way

Describe the pattern if examples mislead.

- ✗ A. Never; examples are always required: not true
- ✗ B. When the example matches the rules perfectly: then keep it
- ✗ C. When you want results to copy the example: then keep it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)